

The re-humanisation of L6, L9 and L10: opportunistic re-colonisation of *Homo sapiens* by an animal-adapted *Mycobacterium tuberculosis* complex lineage, without genomic reversion

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Abstract

Three lineages of the *Mycobacterium tuberculosis* complex (MTBC) – L6, L9 and L10 – sit phylogenetically inside a predominantly animal-adapted clade yet are strictly human-adapted in the extant record. We asked whether this re-humanisation proceeded by genomic reversion of the animal-clade pseudogenisation load or by opportunistic retention. From 1 556 L6, 39 L9 and 3 L10 genomes, plus a stratified 1 849-strain background, we computed the core variant set of each re-humanised lineage and subtracted the cores of every other clade. The resulting signature contains 49 single-position events exclusive to $L6 \cap L9 \cap L10$, including three high-impact loss-of-function events in *Rv0909*, *Rv0386* and *PPE30*, and 27 missense events in genes such as *eccE1*, *cydB*, *wvrA*, *obg* and *phoH2*. The signature is robust to threshold choice (48 / 49 SPDIs invariant from 80% to 95%) and none of the 49 is ancestral when polarised against *M. canettii*. A strict branch decomposition isolates a 23-SPDI core fixed across all three lineages and a 71-SPDI synapomorphy set on the (L6, L9) branch absent from L10 (against 1 and 0 on alternative pairwise branches), supporting an internal topology (L10, (L6, L9)). Crucially, none of the 48 high-impact animal-clade synapomorphies is reverted in any re-humanised lineage. A McDonald–Kreitman test fails to detect an excess of non-synonymous fixations ($d_N/d_S = 0.89$, $p = 0.42$), and a null-permutation control shows that the nominal ESX and PE/PPE enrichments reflect mutational exposure rather than positive selection. The re-humanisation is therefore *additive* and *passive*, not regressive nor adaptive. Tip-dated inference places the crown TMRCA of (L6, L9, L10) at ~ 992 BCE and its divergence from the Wildlife clade at ~ 1501 BCE, and an ancestral character reconstruction localises a single internal *animal* $\rightarrow$ *human* transition on the stem branch – ruling out three independent re-humanisations. We propose that these three lineages represent precarious, context-dependent anthroponoses rather than broadly transmissible modern ecotypes comparable to L2 or L4.

1 Introduction

The *Mycobacterium tuberculosis* complex (MTBC) is a clonal group of obligate pathogens whose host range spans *Homo sapiens* and a striking diversity of mammalian species [Brites et al., 2018, Gagneux, 2018]. Global lineages L1–L10 are conventionally associated

with human tuberculosis, whereas the animal-adapted ecotypes – *M. bovis*, *M. caprae*, *M. microti*, *M. orygis*, *M. pinnipedii*, *M. mungi*, *M. suricattae*, *M. dassie bacillus* and the *chimpanzee bacillus* – form a monophyletic group characterised by distinctive Region-of-Difference (RD) deletion profiles and an accelerated rate of pseudogenisation relative to the modern human lineages [Brites et al., 2018, Zwyrer et al., 2021].

Recent genome-wide phylogenies resolve the MTBC topology after L8 as follows (fig. 1) [Coscolla et al., 2021, Ngabonziza et al., 2020]: a human-strict clade (L1, (L7, (L3, (L2, L4)))) diverges first, and is sister to a grand clade (L5, (Reservoir clade, Specialist clade)). Within the grand clade, the basal L5 retains a purely human host range, while the Reservoir clade comprises all reservoir-competent animal ecotypes capable of unidirectional animal-to-human spillover (*M. bovis*, *M. caprae*, *M. microti*, *M. orygis*, *M. pinnipedii* and the livestock-associated La4). Its sister clade — which we refer to here as the Specialist clade, by contrast with the host-generalist, reservoir-plus-spillover regime of the Reservoir clade — gathers lineages that are each strictly confined to a single host type. The Specialist clade is itself bipartite: on one side, a quartet of purely animal lineages that we refer to here as the Wildlife clade (*chimpanzee bacillus*, *M. mungi*, *M. dassie*, *M. suricattae*); on the other, the three lineages L6, L9 and L10, all of which are strictly human-adapted in extant surveillance.

This topology raises a specific evolutionary paradox. The common ancestor of the Specialist clade must already have carried the reductive-evolution signature that characterises the animal-adapted clade as a whole – pseudogenisations of *mmpL12*, *pks15*, *fadD11*, the PPE5–8 cluster, *cobK/cobL* and related genes [Brites et al., 2018]. The Wildlife clade quartet specialised on non-human hosts. How, then, do L6, L9 and L10 occupy a strictly human niche? Two non-exclusive hypotheses can be formulated. Under the *reversion* hypothesis, the re-humanised branches restored the pre-adaptation-free ancestral state: one expects back-mutations or compensatory loss of animal-clade pseudogenes. Under the *opportunistic* hypothesis, the re-humanised branches inherited the animal-adaptation load unchanged and colonised *H. sapiens* through new, additional modifications on top of it.

Beyond the phylogenetic curiosity, these hypotheses have concrete epidemiological consequences. L6 is the dominant cause of tuberculosis in parts of The Gambia, Guinea-Bissau and Sierra Leone [de Jong et al., 2010, Silva et al., 2022], where it co-circulates with L5 and with modern lineages. L9 and L10 are far rarer, the former restricted to the Horn of Africa (Djibouti / Somalia) [Coscolla et al., 2021] and the latter, very recently defined, known from a handful of Central African isolates. Their genomic trajectories are thus of direct relevance to understanding why human-adapted tuberculosis in parts of Africa still carries a conserved animal-clade-style reductive signature.

This article investigates the genomic signature of the three re-humanised lineages in parallel and tests the reversion-versus-opportunism dichotomy formally. We first describe the host-range distribution across the whole grand clade (section 3.1), confirming the unidirectional nature of Reservoir-clade spillovers and the strict human confinement of L6, L9 and L10. We then derive a signature of 49 variants exclusive to the three re-humanised lineages (section 3.2) and show that none of the 48 animal-clade high-impact synapomorphies are reverted in L6, L9 or L10 (section 3.4). We conclude with functional enrichment (section 3.5), preliminary selection (section 3.6), dating (section 3.7) and ancestral host reconstruction (section 3.8) analyses, and discuss the re-humanisation as an opportunistic, additive and ecologically precarious event (section 4).

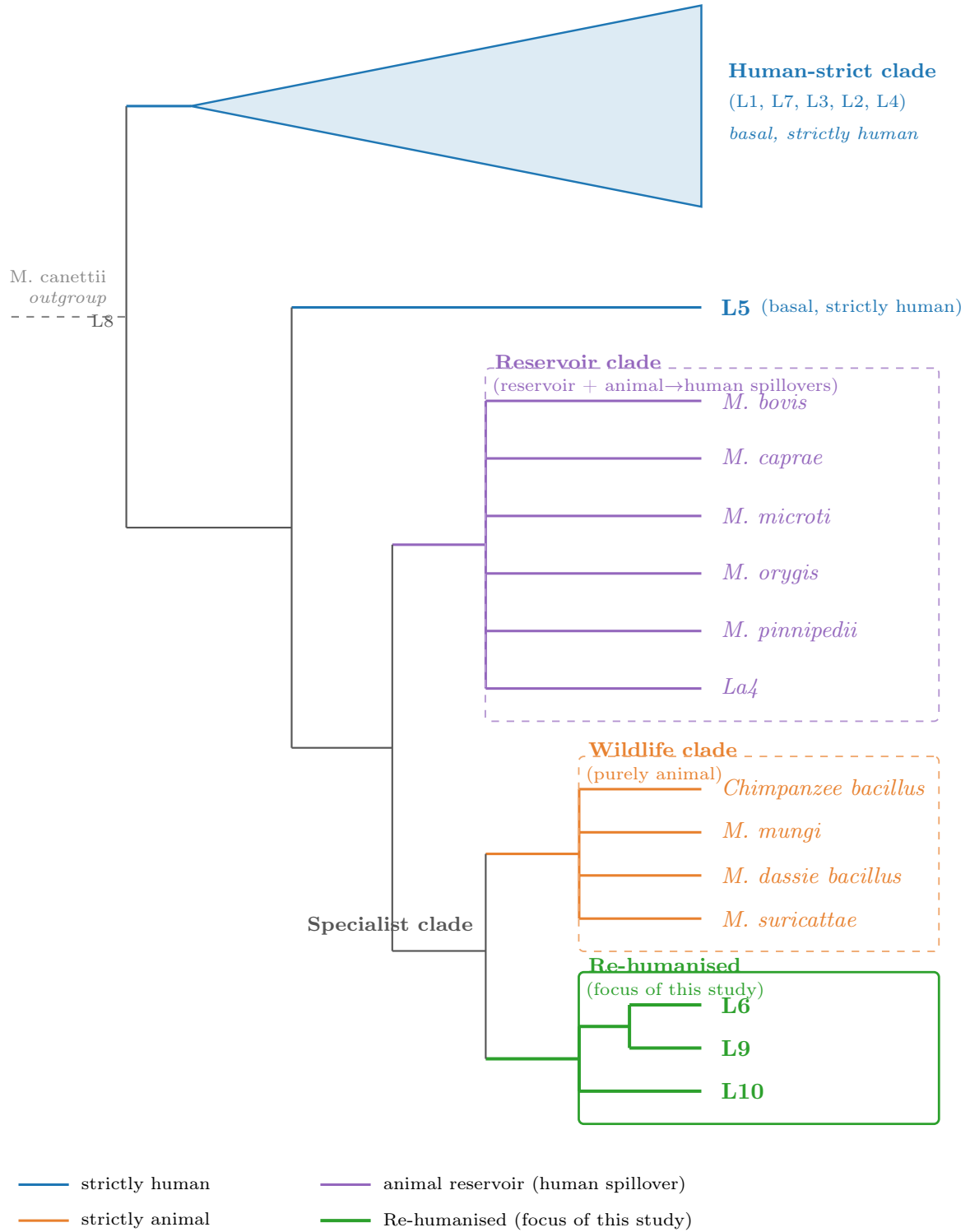


Figure 1: Topology of the MTBC after the L8 divergence (schematic, not to scale). The human-strict clade (L1, L7, L3, L2, L4) diverges first from a grand clade split into L5 (strictly human, basal), Reservoir clade (animal reservoir with unidirectional animal-to-human spillovers: *M. bovis*, *M. caprae*, *M. microti*, *M. orygis*, *M. pinnipedii*, *La4*), and the Specialist clade (itself split into the Wildlife clade (purely animal), and the re-humanised L6/L9/L10 lineages, boxed). The internal topology of the re-humanised crown is resolved as (L10, (L6, L9)) on the basis of the strict 71:1:0 synapomorphy asymmetry across pairwise branches (section 3.3, table 4); the (L6, L9) common branch carries 71 strict synapomorphies and 124 quasi-fixed synapomorphies absent from L10. Branch colour encodes the extant host niche; *M. canettii* serves as the implicit outgroup.

2 Materials and Methods

2.1 Dataset

All genomes analysed in this study were retrieved from TBannotator, a high-throughput annotation platform that maintains a curated catalogue of MTBC genomes downloaded from the NCBI Sequence Read Archive [Guyeux et al., 2024]. All analyses below operate on a frozen August 2024 snapshot of TBannotator (136 774 strains) to guarantee that the L6/L9/L10 cohorts and the comparison clades are drawn from identical source data and that the 96 004-strain classified pool described below is reproducible. For every accession, the platform executes a standardised short-read workflow: quality control with fastp [Chen et al., 2018], alignment to the H37Rv reference (NC_000962.3, 4.41 Mb; Kapopoulou et al., 2011) with BWA-MEM [Li and Durbin, 2009], variant calling with Freebayes [Garrison and Marth, 2012], and functional annotation with SnpEff [Cingolani et al., 2012]. All variants are normalised to SPDI notation (Sequence–Position–Deletion–Insertion), the canonical NCBI format that eliminates the alignment ambiguities of raw VCF records. A curated local catalogue stores, per strain, the SPDI table, region-of-difference profile, IS6110 insertion map and spoligotype.

Strains admitted to TBannotator pass three automated quality gates: GC content within 61–69%, breadth of coverage of the H37Rv reference $\geq 95\%$ (fraction of reference positions covered by at least one read), and mean sequencing depth $\geq 15\times$. Identical sequences, identified by the `-parse` module of RAXML-NG [Kozlov et al., 2019], are relocated to an exclusion archive. These filters remove approximately 11.6% of candidates and yield 96 004 classified strains spanning 73 lineages. Classification is carried out in parallel through five published SNP-barcode systems: Coll [Coll et al., 2014], Stucki [Stucki et al., 2016], Shitikov [Shitikov et al., 2017, 2023], Napier [Napier et al., 2020] and Freschi [Freschi et al., 2021], together with an internal refinement system through which Lineage 10 was originally identified [Guyeux et al., 2024].

From this resource, we sampled the re-humanised lineages exhaustively: L6 contributes 1 556 strains (distributed across sub-lineages L6.1.1 = 936, L6.1.2 = 256 and L6.2 = 364); L9 contributes 39 strains; and L10 contributes 3 strains. Among the L10, two (ERR2707158 and SRR29230781) carry essentially identical SPDI haplotypes (intersection $\geq 99.9\%$) and therefore contribute as a single phylogenetic unit, leaving an effective L10 sample of two unique haplotypes.

The L6 sub-lineage partition extends Coscolla et al. [2021] by introducing an intermediate node, L6.1, that recovers two of Coscolla’s three equivalent-rank sub-lineages (corresponding to our L6.1.1 and L6.1.2) as sister clades sharing a common ancestor distinct from the third (our L6.2). The L6.1 node is supported by 68 strict synapomorphies (variants present at $\geq 90\%$ in both L6.1.1 and L6.1.2 and absent at $< 10\%$ from L6.2), including a frameshift in *espK* (Rv3879c, p.Thr143fs) that refines the *espK* pseudogenisation initially reported in the L6 reference genome [Bentley et al., 2012, Bold et al., 2012] and often presented in the literature as a lineage-wide marker: in fact, only the L6.1 sub-clade carries the truncation while L6.2 retains an intact copy. The full list of 68 supporting variants is given in Supplementary Table S5. The defining SPDI markers used to assign lineage and ecotype labels throughout this study, on H37Rv coordinates (NC_000962.3), are listed in table 1 for every taxon evoked in this paper.

For comparison clades (L5, Reservoir clade sub-lineages, Wildlife clade quartet and L1–L4, L7 from the human-strict clade), two stratified random samples were drawn from

Table 1: Defining SPDI markers for every lineage and ecotype evoked in this study. Coordinates refer to the H37Rv reference genome NC_000962.3 and the convention is position:reference:alternative. Entries marked [†] originate from the broader internal scheme used by the TBannotator pipeline, where no single defining SPDI is yet established for the parent clade. Indentation reflects the topology of fig. 1.

Lineage / ecotype	Defining SPDI on NC_000962.3
<i>Human-strict clade (post-L8)</i>	
L1	1754298:A:C
L2	798354:G:C
L3	207078:G:C
L4	paraphyletic (see L4 sub-lineages in Coll et al., 2014, Stucki et al., 2016, Freschi et al., 2021)
L7	562310:C:T
<i>Grand clade — basal</i>	
L5	3986986:G:T
<i>Grand clade — Reservoir clade (animal reservoir + spillovers)</i>	
<i>M. bovis</i> (La1) [†]	147872:A:C
<i>M. caprae</i> (La2) [†]	65777:T:G
<i>M. microti</i>	1003763:A:G
<i>M. orygis</i> (La3)	3234183:G:C
<i>M. pinnipedii</i>	1096398:G:A
La4 (livestock-associated)	2964336:C:T
<i>Grand clade — Specialist clade / Wildlife clade (purely animal)</i>	
Chimpanzee bacillus	2168451:T:C
<i>M. mungi</i>	2051439:G:A
<i>M. dassie</i> / <i>M. suricattae</i> [†]	1588012:C:T (combined clade)
<i>Grand clade — Specialist clade / re-humanised (focus of this study)</i>	
L6	982362:G:T
L6.1 (intermediate node, this study)	2501400:G:T
L6.1.1 (= Coscolla L6.1)	2180415:C:T
L6.1.2 (= Coscolla L6.2)	2382622:G:C
L6.2 (= Coscolla L6.3)	1028468:G:A
L9	1105922:T:C
L10	1038623:C:A

the local strain catalogue. The grand-clade cohort was sampled with a per-sub-lineage cap of 200 strains applied to the 3 122 strains catalogued in the grand clade (L5, Reservoir clade, Wildlife clade and re-humanised L6/L9/L10 — i.e. all post-L8 lineages except the human-strict clade), yielding an analytical sample of 1 849 strains. The human-strict comparison cohort was sampled separately with a per-major-lineage cap of 30 strains (L4 represented by 30 randomly drawn sub-lineages at two strains each), yielding 294 strains in total — 120 L1, 60 L2, 30 L3, 54 L4 and 30 L7 — which suffice to estimate the L1–L4, L7 core SPDI sets used as outgroups in the signature subtraction described below.

All downstream phylogenetic analyses were performed on a binary pan-SPDI alignment, in which each column records the presence or absence of a given SPDI variant across the dataset. At the typical scale of this study (50 000–200 000 columns $\times$ 50–5 000 taxa), this format is computationally tractable under the BIN+G model in RAxML-NG [Kozlov et al., 2019]; it is also indispensable because classical nucleotide alignment against H37Rv introduces a reference bias that is particularly severe for the divergent lineages analysed here (L5, L6, L9, L10 and animal-adapted ecotypes). Phylogenetic inference has been shown to be robust to the choice among nucleotide substitution models for well-supported clades [Abadi et al., 2019], and we expect the same robustness to extend to the more radical shift from nucleotide to binary encoding when the BIN+G generative model captures the same presence/absence signal that GTR+G would have captured.

2.2 Host range classification

Host metadata were harvested from NCBI through a chained Entrez pipeline (ESearch $\rightarrow$ ESummary $\rightarrow$ EFetch on BioSample and BioProject). Each record was lexically classified as *human*, *animal-expected* (host species matching the known reservoir of that lineage), *animal-other* (any other animal host) or *NA* (no informative metadata). Project titles were deliberately excluded from classification to avoid systematic false positives driven by the keyword “tuberculosis”. Host classification was complemented by RD-barcoding [Coll et al., 2014] using 26 published RD markers, in order to flag potential mis-typings (e.g. L6 strains carrying an animal-clade RD deletion profile).

2.3 Reservoir-versus-dead-end test

For each Reservoir-clade lineage with at least ten human isolates, a Jaccard-based hierarchical clustering of the core variants was used to distinguish unidirectional animal-to-human spillover (human isolates scattered as terminal leaves within animal-majority clusters) from genuine human-to-human transmission (human-only sub-clusters of ≥ 3 isolates). The same clustering was applied to verify that humans in L6, L9 and L10 form the backbone of the tree rather than embedded spillovers.

2.4 Core-genome intersection and signature construction

For each group, the *core* variant set was defined as the set of SPDIs observed at $\geq 90\%$ frequency among the strains of that group. The shared core of the three re-humanised lineages was computed as

$$S_{\text{commun}} = \text{core}(\text{L6}) \cap \text{core}(\text{L9}) \cap \text{core}(\text{L10}), \quad (1)$$

and the exclusive signature as

$$S_{\text{exclusif}} = S_{\text{commun}} \setminus (\text{core}(\text{L5}) \cup \text{core}(\text{Reservoir clade}) \cup \text{core}(\text{Wildlife clade}) \cup \bigcup_{i \in \{1..4,7\}} \text{core}(\text{Li})). \quad (2)$$

The sensitivity of the signature to the core-frequency threshold was assessed by re-deriving S_{exclusif} at thresholds 0.80, 0.85, 0.90 and 0.95, using the same sub-lineage sampling caps and random seed, and by comparing the resulting signatures to one another both with and without additional outgroup polarisation against the *M. canettii* core (fig. S1, table S1).

To resolve the internal architecture of the signature, we then decomposed it by phylogenetic branch under two complementary definitions. A SPDI was assigned to the *stem trio* branch (TRI) when it satisfied a chosen frequency criterion in all three of L6, L9 and L10 simultaneously while being absent from every other clade; to the pairwise branch $LxLy$ when it satisfied the criterion in two of $\{L6, L9, L10\}$ but not in the third; and to a tip apomorphy branch Lx when it was confined to a single lineage. Two stringency levels were applied in parallel: a *strict* definition ($\geq 100\%$ in the included lineages, $\leq 0\%$ in each excluded clade or lineage) which counts only positions that are fixed in every sampled strain of the included groups and entirely absent in every other strain, and a *permissive* definition aligned with the article-level core ($\geq 90\%$ in the included lineages, $\leq 10\%$ in each excluded group). The strict definition isolates the truly invariant fixations of each branch, while the permissive definition recovers the quasi-fixed positions that constitute the article-level signature.

2.5 Functional annotation

Each SPDI of the exclusive signature was annotated with snpEff 5.0 [Cingolani et al., 2012] against the H37Rv database, reporting effect class (HIGH, MODERATE, LOW, MODIFIER), gene name, HGVS.p and functional category. Gene category assignments (ESX / *esx* / *ecc*, PE/PPE, cell wall, lipid metabolism, intermediary metabolism, regulator, hypothetical, other) followed the Tuberculist/Mycobrowser classification [Kapopoulou et al., 2011].

Three complementary layers refined this annotation for the high-impact and missense variants. The domain architecture of the three loss-of-function loci and of the *Rv0909–Rv0910–Rv0911* segment was determined with hmmscan [Eddy, 2011] against the Pfam-A profile-HMM library [Mistry et al., 2021] under family-specific gathering-threshold (`--cut_ga`) cutoffs. The functional impact of every missense substitution was then scored with the ESM-1v protein language model [Meier et al., 2021] as a zero-shot masked-marginal log-likelihood ratio at the mutated residue, $\text{LLR} = \log p(\text{alt} \mid c) - \log p(\text{wt} \mid c)$, where c is the wild-type context with the focal residue masked; substitutions strongly disfavoured by the model return large negative LLR, whereas tolerated substitutions return near-zero values. We report the full LLR distribution rather than a binary call; the heuristic landmark $\text{LLR} < -7.5$ used below to delimit a strongly-disfavoured tail is taken from human-variant benchmarks and is not calibrated for bacterial proteins, so it is used descriptively only. Wild-type protein sequences were taken from the H37Rv NC_000962.3 coding sequences so that residue numbering matched the snpEff HGVS.p annotations, and proteins exceeding the 1 022-residue context window were scored on a window centred on the substitution. Per-residue sparse-autoencoder features from the ESM Atlas [Lin et al., 2023] were additionally consulted as an exploratory, hypothesis-generating description of local biochemical context; being interpretability constructs without a validated mapping to variant effect, they were not used as a quantitative impact predictor.

2.6 Reversion test

The 48 high-impact (frameshift or stop-gained) synapomorphies previously identified as fixed on the ancestral branch of the Specialist clade and inherited by L6 were tested individually for reversion. A variant was scored as a *strict reversion* if its frequency was $\geq 90\%$ in the Wildlife clade and $< 10\%$ in each of L6, L9 and L10 (i.e. lost on the re-humanisation branch after having been shared with the animal-pure sister group).

2.7 Functional enrichment

A hypergeometric test compared the observed number of signature genes in each functional category to the expectation derived from the H37Rv background (ca. 4 008 CDS). p -values were adjusted for the 8 categories tested with the Benjamini–Hochberg procedure. Given the small size of the signature ($n = 49$), enrichment is reported both as nominal p -value and BH-adjusted q -value. Because ESX and PE/PPE gene families are known mutational hotspots in MTBC, the genome-wide hypergeometric test is expected to overestimate their enrichment in any SPDI subset drawn from the intra-clade polymorphic pool. We therefore complemented the hypergeometric analysis with a null-permutation test in which 10 000 sets of 49 SPDIs were drawn uniformly at random from the 1 674 snpEff-annotated polymorphic SPDIs segregating in the pooled L6 $\cup$ L9 $\cup$ L10 sample, and the empirical p -value was computed as the fraction of null draws reaching or exceeding the observed count in each category.

2.8 Selection analysis

The 49-SPDI signature was tested for departure from neutrality using three complementary analyses. First, SPDIs were partitioned into non-synonymous (N : missense and high-impact loss-of-function) and synonymous (S) classes from the snpEff annotation, and compared to the neutral expectation derived from the Nei–Gojobori decomposition of H37Rv [Nei and Gojobori, 1986]. The H37Rv reference yields S -site count = 1 034 424 and N -site count = 2 912 178, so $p_{\text{syn}} = 0.2621$ and the neutral $N:S$ ratio is 2.82. Departure was assessed with a two-sided exact binomial test on the observed number of N events given the coding total. Second, a McDonald–Kreitman contingency table [McDonald and Kreitman, 1991] contrasted the $N:S$ ratio of the signature (divergence) with the $N:S$ ratio of the intra-clade polymorphism (SPDIs segregating at 10–90% frequency in the pooled L6 $\cup$ L9 $\cup$ L10 sample of 383 strains). The Neutrality Index was computed as $\text{NI} = (P_n/P_s)/(D_n/D_s)$, and the Direction of Selection statistic $\text{DoS} = D_n/(D_n + D_s) - P_n/(P_n + P_s)$ [Stoletzki and Eyre-Walker, 2011], with a 95% bootstrap confidence interval obtained from 10 000 resamples. Third, to quantify the potential inflation of N fixation caused by the cross-lineage exclusivity filter used to derive the signature, we simulated draws of 42 coding variants under neutrality and tracked how much lower the sharing probability of synonymous variants with sister groups would have to be (bias factor $b = q_S/q_N$) to reproduce the observed N -fraction; the critical bias b_{crit} is the smallest value at which the observed fraction falls within the null distribution.

2.9 Case-study protocols

The four *M. suricattae* isolates of BioProject PRJNA1109295 flagged with `host=Homo sapiens` (SRR29093504, SRR29093838, SRR29096255, SRR29096559) were compared to

three authentic meerkat isolates from BioProject PRJEB9954 [Parsons et al., 2013]. All seven genomes were profiled for 34 diagnostic RDs; Hamming and Jaccard distances to the *Suricattae* centroid were computed, and mapping statistics (coverage, mean depth, mean MAPQ) were inspected to exclude contamination. Three hypotheses were evaluated: lineage mis-typing (H1), genuine meerkat-to-human zoonosis (H2) and contamination (H3). The eight L6.1.2 strains carrying a convergent RDpan deletion (section 2.2) were tested for monophyly by Fitch parsimony on a sub-tree of L6.1.2 inferred from the L6 sample assembled here.

2.10 Molecular dating

A stratified sub-sample of 99 taxa was assembled for tip-dated phylogenetic inference: 50 L6 genomes drawn proportionally from the three L6 sub-lineages (936 L6.1.1, 256 L6.1.2 and 364 L6.2), 21 of the 39 L9 strains (those with collection-date metadata available at the time of inference), all 3 L10 strains (including the two near-identical haplotypes flagged in §2.1), 10 outgroup taxa from the Reservoir clade (stratified across *M. bovis*, *M. caprae*, *M. orygis*, *M. pinnipedii* and *M. microti*), 9 Wildlife clade taxa (*M. dassie*, the three authentic *M. suricattae* meerkat isolates ERR970409–ERR970412 from Parsons et al. [2013] with the four PRJNA1109295 suspects excluded, *chimpanzee bacillus* and *M. mungi*), 5 L5 genomes and H37Rv as outgroup. An SNP alignment of 17 385 variable positions was extracted from the per-strain SPDI files, excluding invariant columns. Maximum-likelihood inference was performed with IQ-TREE 2 [Minh et al., 2020] under the GTR+ASC+G model (ascertainment-bias correction for SNP-only data), with ultrafast bootstrap (1 000 replicates) and SH-aLRT branch supports. Tip-dating was then carried out with TreeTime v0.11.4 [Sagulenko et al., 2018] using a strict clock, collection-date calibrations parsed from SRA metadata (modern strains in the interval [2010:2022], H37Rv fixed at 1998) and the MTBC consensus genome-wide clock rate of 1×10^{-8} substitutions per site per year [Menardo et al., 2019], rescaled to 2.54×10^{-6} per variable-site per year for the 17 385-SNP alignment ($1 \times 10^{-8} \times 4.41 \times 10^6 / 17\,385$); the rescaling is critical, as the unrescaled rate produces pathological TMRCAs. Ninety-percent marginal-posterior confidence intervals were obtained from TreeTime for nodes resolved in the main auspice output; for clades collapsed into internal polytomies, dates and intervals were propagated from descendant tip dates on the nexus time-tree and are flagged as coarse. Deep-node estimates were cross-checked against the published MTBC root ages of Comas et al. [2013] and Bos et al. [2014]. The sensitivity of the date estimates to the clock rate prior was assessed by replicating the TreeTime analysis with three alternative rates spanning an order of magnitude: a low rate (5×10^{-9} sub/site/yr, aDNA-calibrated sensu Bos et al., 2014), the reference rate used here (1×10^{-8} sub/site/yr, Menardo et al., 2019) and a high rate (5×10^{-8} sub/site/yr, short-term estimate sensu Eldholm et al., 2016), each rescaled identically to the 17 385-SNP alignment (fig. S2, table S2).

2.11 Epidemic success quantification (THD)

Phylogenetic clustering of each lineage was quantified with the Time-scaled Haplotypic Density (THD) introduced by Rasigade et al. [2017]. Because a calibrated tip-dated tree is available, we applied the time-domain formulation: for each tip i , $\text{THD}_i = (n - 1)^{-1} \sum_{j \neq i} K(d_{ij}/\text{bw})$, where d_{ij} is the patristic temporal distance on the TreeTime output (years) and K is the standard-normal Gaussian kernel. Pairwise patristic times on the

99-taxon timetree range from 11 to 9 331 years (median 6 022 years), so the bandwidths originally tabulated by Rasigade et al. (2/5/10 y, calibrated on endemic transmission chains) collapse the kernel to machine zero for nearly every tip. We therefore centred the analysis on $bw = 200$ y (Rasigade’s t_{50} endemic timescale) and bracketed it with 50 y and 500 y as a sensitivity envelope. Pre-declared two-sided Mann–Whitney U tests with Bonferroni correction across 11 contrasts per bandwidth were used to compare groups (L6, L9, L10, pooled re-humanised, Reservoir clade, Wildlife clade, L5). Full outputs are given in fig. S3 and table S3.

2.12 Ancestral host state reconstruction

Ancestral character reconstruction of host state was performed with PastML v1.9 [Ishikawa et al., 2019] on the 99-taxon tip-dated tree of section 2.10, using the marginal posterior probabilities (MPPA) prediction method under the F81 substitution model with two states (*human*, *animal*). State frequencies were estimated by maximum likelihood, yielding $\pi_{\text{human}} = 0.685$ and $\pi_{\text{animal}} = 0.315$ (log-likelihood -11.84 , scaling factor 7.6×10^{-5}). Of the 99 tips, 22 carried a curated *human* host label, 11 a curated *animal* label, and 66 were of unknown host (no informative SRA metadata). *Unknown* tips were treated as missing data (empty cell in the input TSV), so that PastML inferred their state from the topology over a two-state model rather than introducing a spurious third state. An exploratory run that encoded *unknown* as a third symbol produced obvious artefacts (e.g. the L6 crown reconstructed at $P = 0.90$ *unknown*) and was discarded. Posterior probabilities were extracted at five narrative nodes (MTBC root, Reservoir-clade ancestor, stem of the (Wildlife clade, (L6+L9+L10)) split, crown of (L6, L9, L10), Wildlife-clade ancestor) and at the three re-humanised crowns. The minimum number of host-state transitions on the tree was counted by Fitch parsimony from the maximum-likelihood ancestral states.

3 Results

3.1 Host range across the grand clade

We classified the host of 1 849 strains sampled from the grand clade plus a comparison of $\sim 1\,000$ L5 genomes (table 2). Every re-humanised lineage is strictly human: 340 / 340 L6 with informative host metadata (plus a single documented cat spillover from a human contact in Italy, Alba et al., 2025), 21 / 21 L9 and 2 / 2 L10. L5 is similarly pure-human (183 / 183). The Wildlife clade quartet is pure-animal for *M. chimpanzee*, *M. mungi* and *M. dassie*; the four *M. suricattae* records with **host=Homo sapiens** in the WHO CPATH catalogue (PRJNA1109295) are suspected lineage mis-typings and are analysed in detail in section 3.9.

The Reservoir clade shows the inverse pattern: animal reservoirs with unidirectional human spillovers. In *M. bovis* (14.4% human) and *M. pinnipedii* (25.0% human among 4 known hosts), the Jaccard clustering places human isolates as terminal leaves embedded within animal clusters, consistent with isolated spillovers. *M. caprae* (14.4% human) shows two multi-human sub-clusters in Spain and Germany, indicating that some spillovers initiate short-range human-to-human chains. *M. orygis* appears anomalously human-heavy at 78.9%, but this figure reflects WGS sampling bias: five multi-human clusters collapse onto five targeted “orygis-in-humans” BioProjects. No Reservoir-clade

lineage shows a human population forming the backbone of its phylogeny the way L2 or L4 do in the modern human clade.

Table 2: Host range across the MTBC grand clade. *n total*: total strains in the curated catalogue; *human*, *animal-exp.*, *animal-oth.*: per-host classification of the metadata-resolved subset; *% hum.*: fraction of known-host records that are human. Verdict follows the Jaccard clustering of section 2.3.

Group	n total	human	animal-exp.	animal-oth.	% hum.	Verdict
L5	662	183	0	0	100.0	pure human
L6	1556	340	0	1	99.7	pure human
L9	39	21	0	0	100.0	pure human
L10	3	2	0	0	100.0	pure human
Pinnipedii	15	1	3	0	25.0	animal + spillover
Microti	33	2	0	0	100.0	animal (biased NA)
Orygis	358	116	10	21	78.9	mixed (WGS biased)
Bovis	123	13	69	8	14.4	animal + spillover
Caprae	281	38	84	141	14.4	animal + partial bi-dir.
La4	21	3	0	4	42.9	mixed, small sample
Chimpanze	1	0	1	0	0.0	pure animal
Mungi	1	0	1	0	0.0	pure animal
Dassie	6	0	4	0	0.0	pure animal
Suricattae	5	4	0	0	100.0*	catalogue anomaly

3.2 A shared core signature of 49 SPDIs exclusive to L6, L9 and L10

The core variant sets sized 2 143 (L6), 2 214 (L9) and 1 837 (L10) SPDIs. Their intersection contains 1 356 variants. Subtracting the cores of L5, of all six Reservoir clade sub-lineages, of the four Wildlife clade lineages and of L1–L4, L7 leaves a signature of 49 exclusive SPDIs. The progressive exclusion is consistent with the phylogenetic distance to $L6 \cap L9 \cap L10$: Wildlife clade members are the closest competitors (Chimpanze retains 1 302 of the 1 356 common variants), and L4 the most distant (298). A sensitivity analysis across core-frequency thresholds ranging from 0.80 to 0.95 returns signatures of size 50, 49, 49 and 48 SPDIs respectively, with 48 of the 49 reference SPDIs forming an invariant core across all four thresholds; a single in-frame insertion in *Rv0230c* (*php*) is lost at the most stringent 0.95 cut-off, and a single missense variant in *Rv1266c* (*pknH*) is gained at the most permissive 0.80 cut-off, leaving all three high-impact loss-of-function events and the eight functionally prominent missense events (*eccE1*, *cydB*, *wvrA*, *obg*, *phoH2* and the three PE/PPE missense changes) inside the invariant core at the article-level $\geq 90\%$ threshold (fig. S1, table S1). Outgroup polarisation against the *M. canettii* core does not remove any of the 49 SPDIs at any threshold, confirming that none of them is ancestral to the MTBC and that all are post-divergence synapomorphies of the re-humanised clade. A second sensitivity test, replacing the human-strict outgroup sample with an L4-dominant strategy proportional to real MTBC human-clade diversity (200 L4 across 17 canonical sub-lineages, 100 L2 covering proto-Beijing and modern Beijing, 30 each for L1, L3 and L7; 371 strains in total), returns the same signature of 49 SPDIs with full preservation

of the reference set (49/49). The signature is therefore robust both to the core-frequency threshold and to the outgroup sampling strategy.

Of these 49 variants, three are high-impact loss-of-function events (table 3). *PPE30* p.Gln162* is a stop-gained event shared with no animal lineage and adds one more pseudogenisation to the ~ 40 PE/PPE disruptions inherited from the Specialist-clade ancestor; the stop codon falls at the C-terminal boundary of the conserved N-terminal PPE domain (Pfam PF00823, residues 2–164) and excises the entire C-terminal two-thirds of the 463-residue protein, including the PPE-SVP subfamily region (Pfam PF12484, residues 308–379), so the truncation is expected to abolish the exported full-length form. *Rv0386*, a 1085-residue LuxR/GerE-family transcriptional regulator carrying a winged helix-turn-helix domain (Pfam PF25872, residues 437–512) and a C-terminal GerE/LuxR DNA-binding HTH domain (Pfam PF00196, residues 1025–1080), carries an early frameshift at p.Val469fs that disrupts the winged-HTH and abolishes the downstream DNA-binding output domain; and *Rv0909* (the antitoxin component – a Pfam MT0933-like antitoxin, PF14013 – of the mycobacteria-specific Rv0909–Rv0910 type II toxin–antitoxin module of Ramage et al., 2009) carries an early frameshift at p.Asn9fs that removes essentially the whole 59-residue protein. Domain assignments were confirmed against Pfam-A with hmmscan under family-specific gathering-threshold cutoffs [Mistry et al., 2021, Eddy, 2011]; an exploratory ESM Atlas sparse-autoencoder read [Lin et al., 2023] of wild-type *Rv0909* additionally suggested amphipathic, ALPS-like (Amphipathic Lipid-Packing Sensor) membrane-associated features rather than a soluble DNA-binding fold, which, although consistent with the atypical character of this module, is reported here as a hypothesis only. Twenty-seven variants are missense changes, including *eccE1* p.Gly346Ala in the ESX-1 translocon, *cydB* p.Met98Ile (cytochrome *bd* oxidase), *uvrA* p.Ala938Ser (nucleotide-excision repair), *obg* p.Ser149Pro (stringent-response GTPase), *phoH2* p.Ala393Val and three PE/PPE missense events (*PPE8*, *PPE29*, *PE_PGRS26*). The signature is completed by one conservative in-frame insertion in *Rv0230c* (*php*), 12 synonymous and 6 upstream variants compatible with neutral drift.

Table 3: Three high-impact loss-of-function events exclusive to $L6 \cap L9 \cap L10$.

SPDI	Gene	Effect	HGVS.p
NC_000962.3:1014703:GA:G	<i>Rv0909</i>	frameshift	p.Asn9fs
NC_000962.3:2043866:C:T	<i>PPE30</i>	stop_gained	p.Gln162*
NC_000962.3:464814:GT:G	<i>Rv0386</i>	frameshift	p.Val469fs

3.3 Branch decomposition of the signature and internal topology of the re-humanised clade

The article-level signature ($\geq 90\%$ in each re-humanised lineage, with strict exclusion elsewhere) treats L6, L9 and L10 as a single block. Because the stem-to-crown latent interval inferred in section 3.7 suggests a non-trivial pre-radiation history, we tested whether the 49 SPDIs are uniformly distributed on the crown branch of (L6, L9, L10) or whether part of them concentrates on a more recent internal branch. We re-derived the signature under the strict definition described in section 2.4 (100% in the included lineages, 0% outside) and partitioned the resulting positions into a stem-trio category (TRI), three

pairwise categories (L6L9, L6L10, L9L10) and three tip-apomorphy categories (L6, L9, L10).

The strict decomposition isolates an unambiguous core of 23 SPDIs fixed simultaneously in every sampled L6, L9 and L10 strain and entirely absent from every other clade (table 4). It contains the three high-impact loss-of-function events of table 3 together with *cydB* p.Met98Ile, *obg* p.Ser149Pro and one ESX synonymous change (*eccCa1* p.Thr121Thr); the missense changes *eccE1* p.Gly346Ala, *phoH2* p.Ala393Val, *uvrA* p.Ala938Ser, *PPE29* p.Ala147Ser and *PE_PGRS26* p.Gln78His enter the trio only when the inclusion criterion is relaxed from 1.00 to 0.90 (i.e. they are quasi-fixed in the three lineages, with intra-lineage residual polymorphism between 90% and 99% rather than complete fixation). At the permissive 0.90/0.10 cut-off, the stem trio expands to 48 SPDIs, recovering the eight functionally prominent missense events listed in section 3.2. The 25 SPDIs gained between the two stringency levels therefore represent quasi-fixations rather than true invariant positions of the trio; we describe the article-level signature as comprising a 23-SPDI *strict core* and a 25-SPDI *quasi-fixed shell*, both inherited at the (L6, L9, L10) crown.

The strict decomposition also resolves the internal branching order of the three re-humanised lineages, which the tip-dated tree of section 3.7 leaves close to a polytomy. We observe a strong asymmetry across the three pairwise categories: 71 SPDIs are strictly fixed in both L6 and L9 while being entirely absent from L10 (and from every comparison clade), against 1 SPDI on the L6L10 branch and 0 on the L9L10 branch (table 4). The L6L9:L6L10:L9L10 ratio of 71:1:0 at the strict cut-off, and of 124:1:0 at the permissive cut-off, supports an internal topology

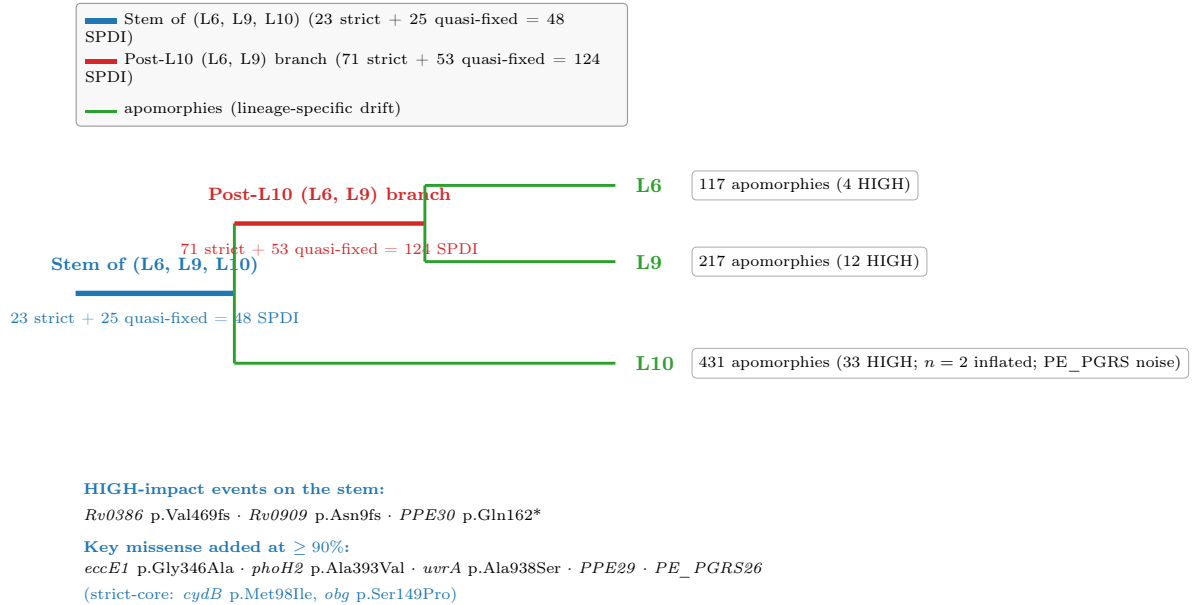
$$(\text{L10}, (\text{L6}, \text{L9})), \quad (3)$$

in which L10 diverges first and L6 and L9 share a more recent common ancestor. The 71-SPDI L6L9 branch carries 2 high-impact pseudogenisations (*Rv0585c* p.Ser518fs, *htpG* p.Gln109*) and three missense changes in ESX-2/3/5 paralogues (*esxH* p.Ala71Ser, *eccD5* p.Pro220Leu, *eccA2* p.Asn543Asp); none of these is shared with L10. We note that the 71:1:0 asymmetry, while suggestive of internal topology (L10, (L6, L9)), is interpreted in the context of the L10 sample size ($n = 2$ unique haplotypes after deduplication), which inflates within-L10 apomorphy counts and could in principle also reduce the count of strict synapomorphies on the L6L10 and L9L10 alternative branches; the asymmetry is corroborated however by the topology of the ML tree of section 2.10, where the (L6, L9) pair is recovered with maximal ultrafast-bootstrap support and the L10 branch is shorter than the inferred stem.

The decomposition thus refines the picture of the re-humanisation event in three respects (fig. 2): (i) only 23 SPDIs (out of the 49 article-level signature) are strictly fixed at the (L6, L9, L10) crown, the remaining 25 being quasi-fixations consistent with shared ancestry but with residual within-lineage polymorphism; (ii) a sizeable additional set of 71 strict synapomorphies separates L10 from the (L6, L9) pair, indicating that a substantial fraction of the genomic divergence between the three lineages is post-trifurcation rather than ancestral; (iii) the apomorphic counts on the three terminal branches (117 in L6, 217 in L9, 431 in L10) are markedly asymmetric, but should be interpreted with caution because the L10 sample contains only $n = 2$ strains, which inflates the count of positions that are necessarily “fixed” at 100% within L10, and because PE/PPE/PE_PGRS variants in L10 are subject to known short-read mapping artefacts in repetitive regions; the L10 figure therefore overestimates the true L10-specific divergence.

Table 4: Branch decomposition of the L6/L9/L10 SPDI signature. Strict counts use inclusion threshold 1.00 and exclusion threshold 0.00; permissive counts use 0.90 / 0.10 in line with the article-level core. “HIGH” counts the frameshift and stop-gained pseudo-genisations within each branch.

Branch	Strict (n)	Permissive (n)	HIGH (strict)	Interpretation
TRI (L6 $\cap$ L9 $\cap$ L10)	23	48	3	stem of the (L6, L9, L10) crown
L6L9 (excl. L10)	71	124	2	supports ((L6, L9), L10)
L6L10 (excl. L9)	1	1	0	alternative (rejected)
L9L10 (excl. L6)	0	0	0	alternative (rejected)
L6 (apomorphies)	117	226	4	lineage-specific drift
L9 (apomorphies)	217	250	12	lineage-specific drift
L10 (apomorphies)	431	447	33	inflated by $n = 2$ sample, PE_PGRS mapping noise



Strict = 100% in included lineages, 0% outside; quasi-fixed = included $\geq 90\%$ but $< 100\%$, all excluded clades $\leq 10\%$.

Figure 2: Stratified view of the L6/L9/L10 SPDI signature. The stem of the (L6, L9, L10) crown carries a 23-SPDI strict core (three high-impact pseudogenisations plus *cydB* and *obg*) and a 25-SPDI quasi-fixed shell (*eccE1* p.Gly346Ala, *phoH2* p.Ala393Val, *uvrA* p.Ala938Ser, *PPE29*, *PE_PGRS26* and others entering only at the article-level $\geq 90\%$ threshold), totalling the 48-SPDI stem trio at the permissive cut-off. The post-L10 (L6, L9) common branch carries an additional 71 strict and 124 quasi-fixed synapomorphies absent from L10, against 1 on the L6L10 alternative and 0 on L9L10 (table 4); this 71:1:0 asymmetry resolves the internal topology as (L10, (L6, L9)). Terminal apomorphy counts are marked next to each tip; the L10 figure is inflated by the $n = 2$ L10 sample and by short-read mapping noise on PE/PPE/PE_PGRS loci.

3.4 No reversion of animal-clade high-impact synapomorphies

The 48 high-impact synapomorphies of the Specialist-clade ancestor (frameshift or stop-gained pseudogenisations in *mmpL12*, *pks15*, *fadD11*, the PPE5–8/16/29–30/45–58 cluster, *cobK/cobL*, *pks3*, *esxC*, *PE_PGRS47*, *senX3*, *pstA1* and others) were tested for reversion in the three re-humanised lineages. Presence at $\geq 90\%$ was observed for 46/48 in L6, 41/48 in L9, 41/48 in L10 and for 37/48 in all three simultaneously; 41/48 in the Wildlife clade and 36/48 in Reservoir clade. No animal-clade synapomorphy satisfies the strict-reversion criterion of $\geq 90\%$ in the Wildlife clade and $< 10\%$ in each of L6, L9 and L10:

$$N_{\text{réversion}} = 0 \quad / \quad 48. \quad (4)$$

The small apparent loss of signal in L9 and L10 (7 SPDIs at intermediate frequencies) reflects the small L10 sample ($n = 2$) and isolated polymorphisms in L9, but no variant crosses the $< 10\%$ threshold required for true back-mutation.

3.5 Functional composition of the signature and null-permutation control

The 49 signature SPDIs distribute across eight functional categories (table 5). A naive hypergeometric test against the H37Rv genomic background (~ 4008 CDS) returns two categories crossing nominal significance without reaching the Benjamini–Hochberg adjusted threshold of $q \leq 0.05$: the ESX / *esx* / *ecc* secretion-system category (5.84-fold nominal enrichment; 2 observed vs 0.34 expected; $p = 0.045$, $q = 0.146$) and the PE/PPE family (2.41-fold; 5 observed vs 2.08 expected; $p = 0.055$, $q = 0.146$). This nominal enrichment is, however, an artefact of the genome-wide reference. ESX and PE/PPE genes are well-known mutational hotspots in the MTBC and are already over-represented in the intra-clade polymorphic pool: of the 1674 snpEff-annotated SPDIs segregating at intermediate frequency in the pooled L6 $\cup$ L9 $\cup$ L10 sample, 34 (2.0%) fall in the ESX / *esx* / *ecc* category against $\sim 0.7\%$ at the genome level, and 230 (13.7%) fall in PE/PPE against $\sim 4.2\%$ at the genome level.

The null-permutation control, which resamples 10 000 sets of 49 SPDIs from this intra-clade polymorphic pool, confirms that the signature is statistically indistinguishable from a random draw in both categories: ESX shows an observed count of 2 against a null expectation of 1.00 ± 0.96 ($p_{\text{emp}} = 0.26$), and PE/PPE shows an observed count of 5 against a null expectation of 6.72 ± 2.36 ($p_{\text{emp}} = 0.83$), the latter value indicating that the signature is in fact marginally depleted in PE/PPE relative to the polymorphic pool. The functional composition of the 49-SPDI signature is therefore compatible with neutral sampling of the ordinary MTBC polymorphism pool, once that pool’s own over-representation of mutation-prone gene families is accounted for. This is consistent with the neutral selection signal reported in section 3.6 and does not exclude locus-specific effects at individual genes such as *eccE1* or *cydB*, which would require gene-level branch-site models or experimental phenotyping to be detected.

The branch decomposition of section 3.3 further mechanistically refines the ESX-family argument. Only one ESX-1 missense, *eccE1* p.Gly346Ala, is fixed on the (L6, L9, L10) stem trio (and only at the permissive $\geq 90\%$ threshold; it is not fixed at 100% in every sampled strain). The other three ESX missense changes that contribute to the nominal 5.84-fold enrichment (*esxH* p.Ala71Ser, *eccD5* p.Pro220Leu, *eccA2* p.Asn543Asp) are not co-fixed with *eccE1* on the stem trio: they are strictly fixed on the post-L10 (L6, L9)

branch and are absent from L10. Conversely, 7 ESX-1 missense changes (*eccB1*, *eccD1* $\times 2$, *eccC2* $\times 2$, *eccA2*, *eccE2*) appear as L10-only apomorphies, where they are inflated by the $n = 2$ sample and the PE_PGRS mapping noise discussed in section 3.3. The signature-level ESX enrichment of table 5 therefore aggregates substitutions that arose on three distinct branches of the re-humanised clade rather than co-fixed on its stem; this stratification is consistent with the neutral null-permutation result and with a stem-only ESX-1 perturbation restricted to a single missense change in *eccE1*.

Table 5: Hypergeometric enrichment of the 49-SPDI signature against the H37Rv background (~ 4008 CDS). Fold = observed / expected; p is nominal, q is Benjamini–Hochberg adjusted across 8 categories.

Category	Bg genes	Obs.	Exp.	Fold	p	q
hypothetical	1000	20	12.2	1.64	0.010	0.082
ESX / <i>esx</i> / <i>ecc</i>	28	2	0.34	5.84	0.045	0.146
PE / PPE	170	5	2.08	2.41	0.055	0.146
other	1260	19	15.4	1.23	0.168	0.337
lipid metabolism	250	1	3.06	0.33	0.958	1.000
intermediary metab.	900	2	11.0	0.18	1.000	1.000
regulator / TF	180	0	2.20	0.00	1.000	1.000
cell wall	220	0	2.69	0.00	1.000	1.000

3.6 Selection signal on the re-humanisation signature

Of the 49 signature SPDIs, 42 fall within annotated coding sequences. Thirty of these are non-synonymous (27 missense plus the three high-impact loss-of-function events of table 3) and twelve are synonymous; the remaining 7 are upstream variants or a conservative in-frame insertion (table 6). The observed $N:S$ ratio of 2.50 (30/12) is lower than the neutral expectation of 31/11 derived from $p_{\text{syn}} = 0.2621$ on H37Rv, and the two-sided exact binomial test confirms no departure from neutrality ($p = 0.71$). The global ratio $d_N/d_S \simeq 0.89$ indicates weak purifying constraint rather than adaptive acceleration.

The McDonald–Kreitman contingency table, assembled with intra-clade polymorphism segregating at 10–90% in the pooled 383-strain $L6 \cup L9 \cup L10$ sample (1382 coding polymorphic sites: $P_n = 894$, $P_s = 488$), yields a non-significant Fisher’s exact test ($p = 0.42$), a neutrality index $NI = 0.73$, and a direction-of-selection estimate $DoS = +0.067$ with 95% bootstrap confidence interval $[-0.076, +0.205]$ that crosses zero. The ascertainment simulation returns a critical bias $b_{\text{crit}} = 1.00$: no differential filtering of synonymous relative to non-synonymous variants is required to reproduce the observed N -fraction under neutrality. We therefore detect *no excess* of non-synonymous fixations on the re-humanisation branch, a result consistent with neutrality but with limited statistical power given the small number of coding sites tested ($n = 42$); the analysis cannot exclude weak adaptive signals at individual loci.

The three analyses converge on the same conclusion: the 49-SPDI signature is statistically indistinguishable from neutral fixation by drift within the three small founder populations of L6, L9 and L10. Taken together with the null-permutation control on the functional composition of the signature (section 3.5), which shows that the nominal ESX and PE/PPE enrichments are already explained by the over-representation of these

Table 6: Two-by-two McDonald–Kreitman table for the 49-SPDI re-humanisation signature. Divergence: variants fixed exclusively on the $L6 \cap L9 \cap L10$ branch. Polymorphism: SPDIIs segregating at 10–90% in the pooled 383-strain sample.

Class	Non-synonymous	Synonymous	Total	% N
Divergence (signature)	30	12	42	71.4
Polymorphism (intra $L6 \cup L9 \cup L10$)	894	488	1 382	64.7

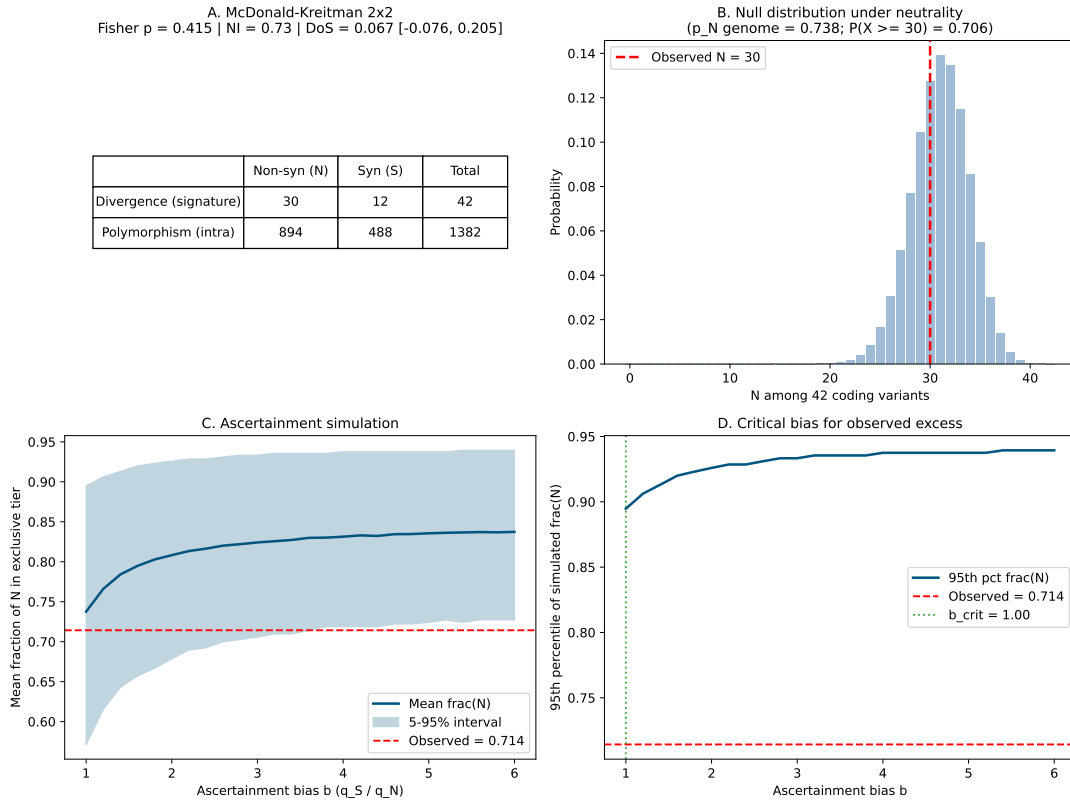


Figure 3: Selection analysis of the 49-SPDI re-humanisation signature. (A) McDonald–Kreitman 2×2 contingency table contrasting signature divergence with intra-clade polymorphism (Fisher’s exact $p = 0.42$, $DoS = +0.067$). (B) Null binomial distribution $P(X = k \mid n = 42, p_N = 0.738)$; the vertical line marks the observed $D_n = 30$ (two-sided $p = 0.71$). (C) Mean N -fraction under the ascertainment simulation as a function of the bias factor b ; the shaded band gives the 5–95% simulation interval. (D) The 95th-percentile envelope intersects the observed N -fraction at $b_{\text{crit}} = 1.00$, i.e. no cross-lineage exclusivity bias is needed to explain the observation.

families in the intra-clade polymorphic pool, this defines a coherent neutral triptych: no aggregate departure from neutrality is detectable at the scale of the signature, and no functional category is enriched beyond the expectation of random draws from the ordinary MTBC polymorphism pool. This does not exclude locus-specific effects at candidate genes such as *eccE1* or *cydB*, which remain testable by gene-level branch-site models or direct phenotyping.

3.7 Timing: single or multiple re-humanisation events?

Tip-dated inference on the 99-taxon SNP alignment (section 2.10) places the MTBC root at 1977 BCE (90% CI [3303 BCE, 738 BCE]). This Late Holocene root estimate is markedly more recent than the ~ 70 kyr coalescence inferred under the out-of-Africa coexpansion model of Comas et al. [2013], but consistent with the post-Neolithic age window favoured by the ancient-DNA anchor of Bos et al. [2014] and with strict-clock estimates that do not enforce a deep human-pathogen codivergence prior. This is the well-known tension between rate-based and aDNA-based MTBC chronologies; the relative ordering of internal nodes used in our subsequent arguments (stem < crown) is robust to this absolute-date uncertainty (section 3.7 sensitivity analysis). The split of the Wildlife clade quartet from the ancestor of the three re-humanised lineages is placed at 1501 BCE [2753 BCE, 449 BCE], and the crown of (L6, L9, L10) at 992 BCE [2081 BCE, 94 BCE]; a summary of the nodes relevant to the re-humanisation question is given in table 7 and the dated tree is shown in fig. 4.

Table 7: Tip-dated TMRCA estimates for MTBC nodes relevant to the re-humanisation of L6, L9 and L10. Intervals in square brackets are 90% marginal posterior from TreeTime; dagger-marked rows (†) are point estimates propagated from descendant tip dates on the time-nexus because the clade is collapsed into an internal polytomy in the auspice output, and therefore lack a proper statistical interval.

Node	TMRCA	90% CI
MTBC root	1977 BCE	[3303 BCE, 738 BCE]
Wildlife clade + (L6 + L9 + L10) stem	1501 BCE	[2753 BCE, 449 BCE]
(L6 + L9 + L10) crown	992 BCE	[2081 BCE, 94 BCE]
L6 crown	174 CE	[507 BCE, 734 CE]
L9 crown [†]	185 CE	point estimate
L10 crown [†]	782 CE	point estimate

Three lines of evidence favour a single ancestral re-humanisation event (H1) over three independent zoonoses (H2). First, (L6, L9, L10) is monophyletic in the ML topology with maximal support (100/100 ultrafast-bootstrap / SH-aLRT), consistent with a single point of descent from an animal-adapted ancestor. Second, the 510-year stem-to-crown gap between the split from the Wildlife clade (1501 BCE) and the (L6, L9, L10) crown (992 BCE), albeit a point-estimate ordering whose 90% CIs overlap substantially ([2753 BCE, 449 BCE] vs [2081 BCE, 94 BCE]), is consistent with a protracted latent phase during which a single re-humanised ancestor existed before radiating into the three observed lineages; we therefore stress that the latent-phase argument rests on the point-estimate ordering and on the topological monophyly of (L6, L9, L10) rather than on a strict statistical separation of the two dates. Third, the crown TMRCA of L6 (174 CE) and

Time-calibrated phylogeny of the re-humanisation panel
(99 tips, IQ-TREE 2 ML + TreeTime strict clock 2.54e-6 sub/SNP-site/yr, MTBC root 1977 BCE)

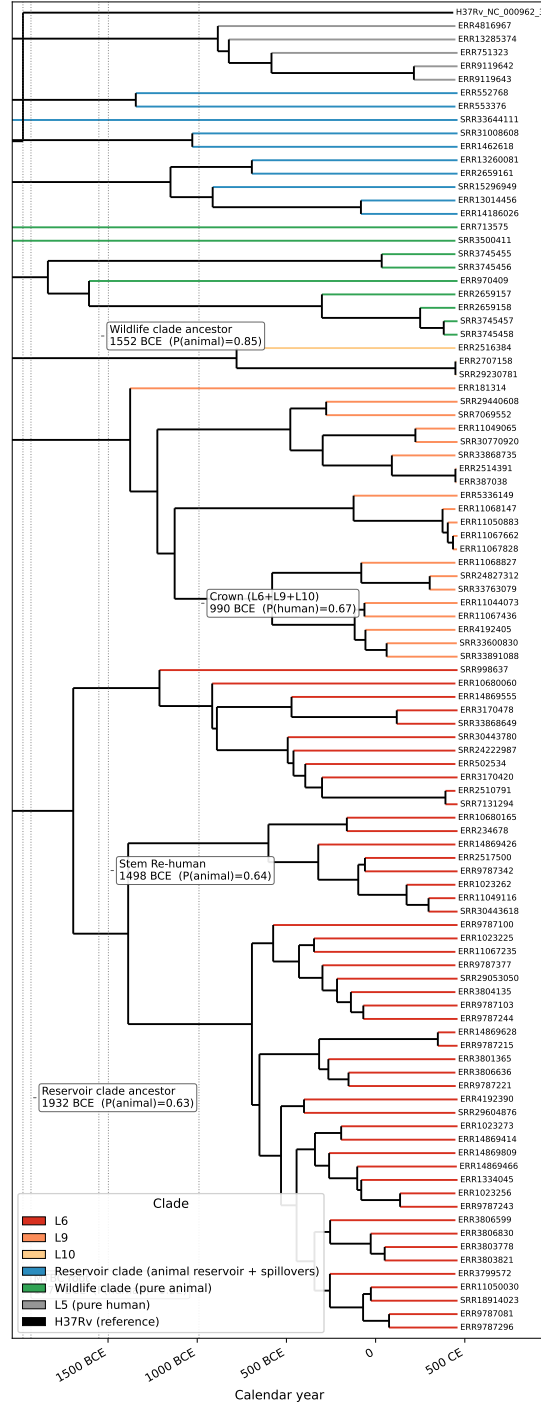


Figure 4: TreeTime dated phylogeny of the 99-taxon re-humanisation panel (IQ-TREE 2 ML topology, GTR+ASC+G, strict clock rescaled to 2.54×10^{-6} per variable-site per year). Branch lengths are proportional to calendar time. The Wildlife clade + (L6 + L9 + L10) stem (1501 BCE) and the (L6, L9, L10) crown (992 BCE) are separated by a ~ 510 -year latent interval. Tip colours identify the four groups (L6, L9, L10, Wildlife clade); Reservoir clade and L5 are outgroup-stratified; H37Rv anchors the root.

L9 (185 CE) are effectively contemporaneous, their 90%-CI envelopes overlap fully, and the later L10 crown (782 CE) still sits inside the monophyletic (L6, L9, L10) clade rather than branching off the Wildlife clade directly; L10 therefore represents a later radiating sister branch of the same re-humanised ancestor, not an independent meerkat-to-human or dassie-to-human spillover.

The dated scenario places the inferred single re-humanisation at the end of the West African Late Bronze Age (~ 1500 BCE), followed by a ~ 500 -year period during which the new lineage did not diversify into the sampled crown clades, and then a first-millennium-CE radiation broadly coincident with the westward expansion of iron-working and the southward Bantu expansion.

To evaluate the sensitivity of this temporal narrative to the strict-clock prior, we replicated the TreeTime analysis with three clock rates spanning an order of magnitude (low 5×10^{-9} , reference 1×10^{-8} and high 5×10^{-8} sub/site/yr; fig. S2, table S2). The monophyly of (L6, L9, L10), the ordering stem < crown < within-lineage crowns, and the positivity of the stem-to-crown gap are invariant across all three rates; the gap itself scales with the rate but remains substantial, from 1016 years at the low rate through 510 years at the reference rate to 102 years at the high rate. Absolute dates scale inversely with the rate and therefore depend directly on the Menardo 2019 prior used here: under the slow aDNA-calibrated rate the re-humanisation would be pushed back into the early Bronze Age, whereas under the short-term fast rate it would occur in Late Antiquity. We therefore report the topological and temporal-ordering conclusions (single ancestral re-humanisation, pre-crown stem, finite latent interval) as robust to clock-rate mis-specification, while flagging the absolute calendar dates as rate-conditional.

Two important caveats qualify this interpretation. The strict-clock assumption is likely to underestimate the true confidence intervals (a relaxed-clock BEAST2 analysis would be required to quantify this), and the CIs for L9, L10, Wildlife clade, Reservoir clade and L5 are derived by tip-date propagation rather than from the marginal posterior, so their absolute values are point estimates reflecting a defect of TreeTime resolution on collapsed polytomies rather than genuine temporal certainty. The clade-level (L6+L9+L10) TMRCA and the Wildlife clade + (L6+L9+L10) stem, which carry proper confidence intervals, are the strongest quantitative support for H1. We also note that under the high-rate prior, the marginal posterior intervals of the stem and crown nodes start to overlap, which weakens the latent-phase argument at that extreme of the rate range.

3.8 Formal ancestral reconstruction of host state

To formally test the post-L8 narrative *human* (L5) $\rightarrow$ *mixed* (Reservoir clade) $\rightarrow$ {*pure animal* (Wildlife clade), *re-humanised* (L6+L9+L10)} and the single-event hypothesis (H1) supported by the topology and timing in section 3.7, we performed an ancestral character reconstruction of host state on the 99-taxon tip-dated tree (section 2.12). The five narrative nodes are recovered at their expected states with posterior probabilities ranging from 0.52 to 0.85 (table 8). The discriminating node – the stem of the split between the Wildlife clade and the re-humanised lineages – is reconstructed as *animal* with $P(\text{animal}) = 0.638$, while the crown of (L6, L9, L10) is reconstructed as *human* with $P(\text{human}) = 0.665$. The three re-humanised crowns are uniformly human ($P(\text{human}) = 0.997$ for L6, 0.990 for L9, 0.880 for L10).

Counting transitions by Fitch parsimony on the maximum-likelihood ancestral states

Table 8: PastML marginal posterior probabilities of the host state (*human*, *animal*) at the five narrative nodes of the MTBC post-L8 topology, under the F81 model with MPPA prediction (log-L = -11.84, $\pi_{\text{human}} = 0.685$, $\pi_{\text{animal}} = 0.315$). The maximum-likelihood (ML) state at each node coincides with the state predicted by the narrative.

Node	ML state	$P(\text{human})$	$P(\text{animal})$
MTBC root	human	0.518	0.482
Reservoir-clade ancestor	animal	0.372	0.628
Stem (Wildlife clade, (L6+L9+L10))	animal	0.362	0.638
Crown (L6+L9+L10)	human	0.665	0.335
Wildlife-clade ancestor	animal	0.154	0.846

yields 1 *human* $\rightarrow$ *animal* event and 5 *animal* $\rightarrow$ *human* events on the 99-taxon tree (table S4). The single *human* $\rightarrow$ *animal* transition occurs on the deep branch leading from the MTBC root to the grand-clade ancestor, consistent with the historical adaptive shift to non-human hosts at the base of the L5+animal+re-humanised clade. Of the five reverse transitions, only one is internal: it is localised on the branch connecting the stem (Wildlife clade, (L6+L9+L10)) to the crown of (L6, L9, L10), exactly the branch on which H1 predicts the unique re-humanisation event. The four remaining *animal* $\rightarrow$ *human* transitions are tip-terminal in the Reservoir clade (Pinnipedii ERR552768, Microti ERR553376, Orygis SRR33644111, Caprae SRR31008608) and correspond to the isolated human spillovers documented in section 3.1; no internal transition is inferred within Reservoir clade, consistent with the reservoir-with-spillover regime. No internal *animal* $\rightarrow$ *human* transition is inferred anywhere within (L6, L9, L10): the three re-humanised crowns are reconstructed as derived from a common already-human ancestor, ruling out a scenario of three convergent zoonoses.

The ancestral host reconstruction therefore formally confirms the post-L8 narrative inferred from the topology and the timing analysis. The single internal *animal* $\rightarrow$ *human* transition predicted by H1 is recovered, and no alternative independent re-humanisation is inferred. Two caveats apply. First, the MTBC root is reconstructed as *human* only with $P = 0.52$, a near-equiprobable signal that reflects the absence of a *M. canettii* outgroup in the tip-dated tree; the out-of-Africa human-origin scenario of Comas et al. [2013] is therefore *compatible* with the reconstruction but not strongly supported by it (Bayes factor ~ 1.08). Second, the four tip-terminal *animal* $\rightarrow$ *human* transitions in the Reservoir clade reflect the deliberate stratified sampling of that clade (10 tips drawn across five sub-lineages to maximise host diversity); a more balanced or larger sample would likely reveal additional spillover events without altering the unique internal animal-to-human transition on the (L6, L9, L10) stem branch. The reconstruction is shown projected onto the tip-dated tree in fig. 5; the full interactive tree is provided as a Supplementary HTML file and the per-node posterior table as table S4.

Ancestral host reconstruction (PastML MPPA + F81) on the 99-taxon tip-dated tree

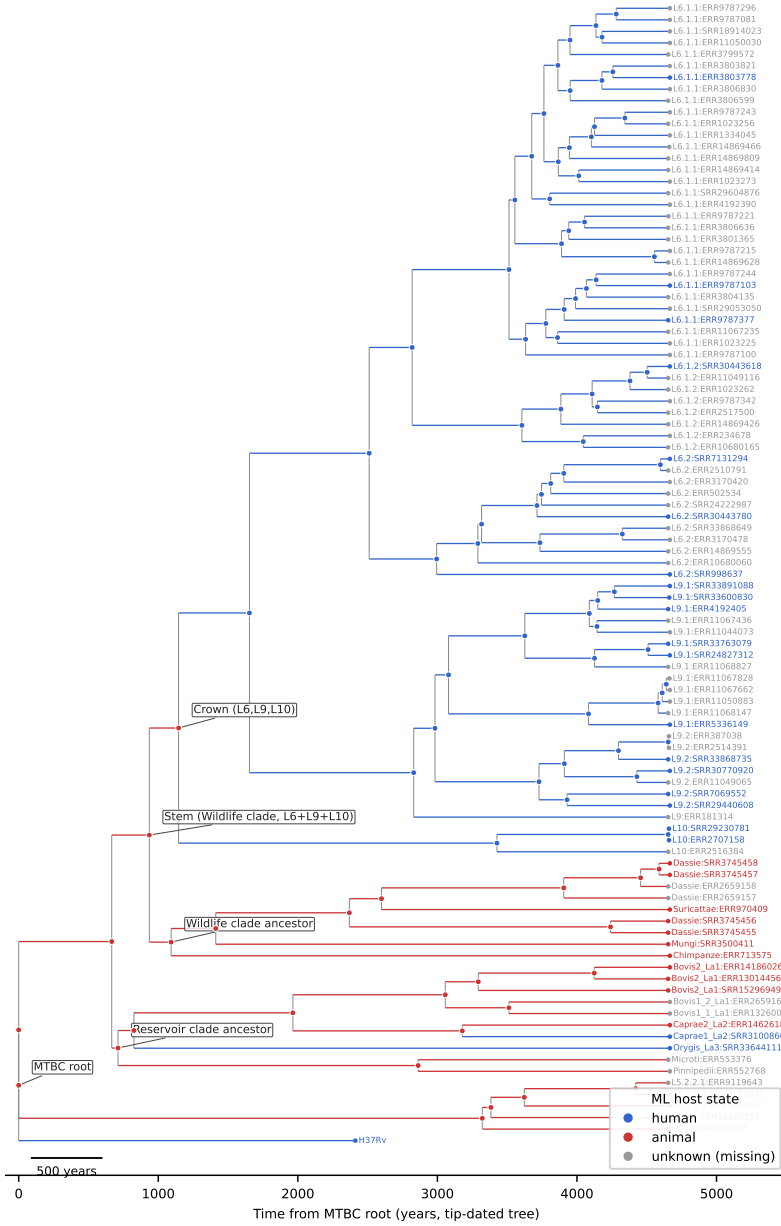


Figure 5: PastML ancestral host reconstruction projected onto the 99-taxon tip-dated tree. Each tip is coloured by its curated host label (human, animal, unknown = missing data); each internal node is coloured by its maximum-likelihood reconstructed state under the F81 model with MPPA prediction. The five narrative nodes (MTBC root; Reservoir-clade ancestor; stem (Wildlife clade, (L6+L9+L10)); crown (L6, L9, L10); Wildlife-clade ancestor) are annotated with their posterior probabilities (table 8). The unique internal *animal* $\rightarrow$ *human* transition predicted by H1 is highlighted on the branch connecting the stem to the crown of (L6+L9+L10); the four tip-terminal *animal* $\rightarrow$ *human* transitions in the Reservoir clade correspond to documented isolated spillovers.

3.9 Case studies: the *Suricattae* catalogue anomaly and the L6.1.2 RDpan sub-clade

PRJNA1109295 *M. suricattae* isolates flagged as human

The four PRJNA1109295 genomes registered with `host=Homo sapiens` (SRR29093504, SRR29093838, SRR29096255, SRR29096559) were profiled against 34 diagnostic RDs and compared to the three authentic meerkat references ERR970409, ERR970410 and ERR970412 from Parsons et al. [2013]. All four suspects exhibit a Hamming distance of zero to the *Suricattae* centroid, sharing the full lineage-specific signature (RDsurRv1799, RDsurppe34, RDsurRv2262, RDsurRv2307c, RDsurRv3324A, RD1das, RDVirsdas and N-RD25das) together with the broader grand-clade deletions inherited from the Wildlife clade branch (fig. 6). Mapping quality is clean (mean depth 52–85, mean MAPQ 59.1, covered bases 97.8–98.2%), ruling out contamination. The SNP-barcoding hypothesis of lineage mis-typing (H1) is incompatible with a complete lineage-specific RD signature, and the contamination hypothesis (H3) is incompatible with the mapping statistics. The genomic evidence is therefore compatible with a zoonotic transmission from meerkats to humans (H2), a mode of transfer not previously reported for *M. suricattae* [Parsons et al., 2013, Dippenaar et al., 2015]. We note however that the isolation metadata in the CPATH / WHO catalogue (isolation source and country missing) leaves open an alternative H1-bis where the submitter entered `Homo sapiens` by default. Confirmation would require contacting the submitter; we therefore flag the four isolates as a documented curiosity rather than a first described meerkat-to-human spillover.

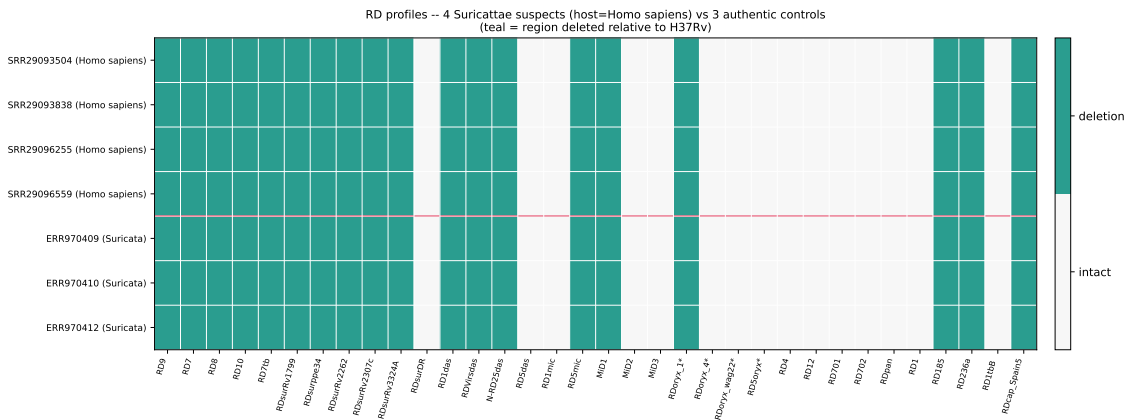


Figure 6: Region-of-Difference profile of the four PRJNA1109295 isolates flagged as `host=Homo sapiens` (top) compared with three authentic *M. suricattae* meerkat isolates from Parsons et al. [2013] (bottom). Dark cells denote RD deletions. The eight *Suricattae*-diagnostic loci (left) are fully shared across suspects and controls.

Convergent RDpan deletion in L6.1.2

Seven of the eight L6.1.2 strains carrying an RDpan deletion (one strain, ERR2517500, was excluded for failing the quality-control criteria of section 2.1) were mapped onto the L6.1.2 sub-tree inferred from the L6 cohort assembled here. Their most recent common ancestor contains 112 tips, 105 of which do not carry the RDpan deletion; the eight targets therefore do not form a monophyletic group. Fitch parsimony returns a minimum of three

independent RDpan-deletion events on the tree, resolving into three pure monophyletic subsets: {ERR3803993, ERR9787060} (patristic distance $< 10^{-5}$), {ERR9787271} in isolation and {ERR3804108, ERR3804120, ERR9787191, ERR9787218} (patristic distances 0.0008–0.002). Inter-cluster patristic distances (~ 0.069) are two orders of magnitude larger, confirming independent origins. No SPDI is shared by all eight strains while being absent from the rest of L6, i.e. no synapomorphy unites them. Clusters 1 and 3 do carry their own exclusive SNP sets (82 and 133 SPDIs respectively), defining two genuine sub-clades within L6.1.2, but these are independent of the RDpan status. RDpan thus behaves as a recurrently-deleted locus within L6.1.2 rather than a clade-defining synapomorphy, consistent with the structural instability of the *pks2*–*pks3* polyketide synthase cluster in which it sits [Bottai et al., 2014]. The convergence echoes, at a genomic scale, the passive and opportunistic mode of re-humanisation: L6.1.2 accumulates further structural losses rather than reversing them, without these losses coalescing into a new host-range outcome.

3.10 Phylogenetic clustering of L6, L9 and L10 relative to established lineages

The precarity of L6, L9 and L10 invoked in the Discussion below is a qualitative claim about census counts and geographic confinement. To test whether it also holds at the level of phylogenetic clustering – i.e. whether, when these lineages do infect a human, they transmit less densely than established human or animal-adapted lineages – we applied the Time-scaled Haplotypic Density metric of Rasigade et al. [2017] to the tip-dated tree (section 2.11). Median THD values at the central bandwidth of 200 years differ by more than forty orders of magnitude across groups (table 9), and the qualitative ranking $L10 > L9 > \text{Wildlife clade} > L6 > L5 > \text{Reservoir clade}$ is invariant across the three bandwidths tested.

Three Mann–Whitney contrasts survive Bonferroni correction at all three bandwidths. The pooled re-humanised cohort is significantly *more* clustered than the Reservoir clade ($p_{\text{Bonf.}} = 7.0 \times 10^{-4}$), as are L6 alone ($p_{\text{Bonf.}} = 2.0 \times 10^{-3}$) and L9 alone ($p_{\text{Bonf.}} = 1.7 \times 10^{-3}$); this is expected, since the Reservoir clade is a paraphyletic stratified sample of five animal sub-lineages deliberately drawn to maximise diversity. Among the re-humanised lineages themselves, L9 is significantly more clustered than L6 ($p_{\text{Bonf.}} = 2.4 \times 10^{-2}$), reflecting the recent radiation of L9.1 (13 of 21 L9 tips) together with the compact L10 triplet. The L6 versus L5 contrast, however, is *not* significant at any bandwidth ($p_{\text{Bonf.}} = 1.00$), and the same holds for the pooled re-humanised cohort against L5; given the small L5 sub-sample ($n = 5$ tips), this is a finding of no detectable difference rather than evidence of equivalence, and the test has limited power to discriminate moderate divergences in clustering. We therefore conclude that we find *no evidence of reduced phylogenetic clustering* in the re-humanised lineages relative to L5; any precarity argument must reside in census count rather than in tree-based transmission density.

Three caveats qualify the interpretation. First, the stratified sub-sampling is unbalanced (50 L6 drawn from 1 556 strains vs 10 Reservoir clade drawn across five sub-lineages), so the absolute L6 value is an upper bound inflated by within-group sampling density. Second, L10 ($n = 2$) and L5 ($n = 5$) are under-powered for Mann–Whitney testing, and the absence of a significant L6–L5 contrast should not be over-interpreted as equality. Third, the canonical THD of Rasigade et al. [2017] was developed for endemic transmission-chain bandwidths of a few years; the 200-year bandwidth used here probes

Table 9: THD median (bandwidth = 200 years) per MTBC group on the 99-taxon tip-dated tree. Rank is by decreasing median. The qualitative ranking is preserved at bw = 50 and 500 years (see table S3).

Group	n	Median THD (bw=200 y)	Rank
L10	3	2.03×10^{-5}	1
L9	21	6.45×10^{-8}	2
Wildlife clade	9	4.00×10^{-9}	3
L6	50	6.63×10^{-13}	4
L5	5	1.39×10^{-28}	5
Reservoir clade	10	9.15×10^{-46}	6

a multi-secular diversification regime that is distinct from, but complementary to, the short-term transmission success for which the metric was originally calibrated.

4 Discussion

4.1 Re-humanisation as opportunistic re-colonisation

The central finding of this study is that the re-humanisation of L6, L9 and L10 did not proceed through reversion of the animal-clade pseudogenisation load. None of the 48 high-impact loss-of-function events that accumulated on the ancestor of the Specialist clade was restored in any of the three re-humanised lineages. Instead, the signature characterising these lineages is additive: three further high-impact pseudogenisations (*Rv0909*, *Rv0386*, *PPE30*) stack on top of the pre-existing animal-clade losses, together with missense changes that fall in gene families associated with phenotypic diversification and antigenic variation in the MTBC [Brites et al., 2018, Gagneux, 2018, Pepperell et al., 2013]. Four independent lines of evidence converge to qualify this additive signature as *passive* rather than adaptive, and to constrain the re-humanisation to a single ancestral event: (i) no reversion of the 48 animal-clade high-impact synapomorphies (section 3.4); (ii) no detectable excess of non-synonymous fixations on the re-humanisation branch, with $d_N/d_S = 0.89$, a McDonald–Kreitman test non-significant at $p = 0.42$ and a direction-of-selection confidence interval crossing zero (section 3.6; absence of detectable signal rather than positive evidence of strict neutrality, as the test has limited power on the 42 coding sites), and no aggregate functional enrichment once mutational exposure is controlled, the null-permutation p -values for ESX ($p_{\text{emp}} = 0.26$) and PE/PPE ($p_{\text{emp}} = 0.83$) showing that the genome-wide nominal signal is an ascertainment artefact of the polymorphic pool rather than a selection signature of re-humanisation (section 3.5); (iii) tip-dated phylogenetic inference, which places the crown of (L6, L9, L10) within a single monophyletic lineage descending from a pre-crown stem, with a positive latent interval consistent with a unique ancestral event rather than three independent zoonoses (section 3.7); and (iv) the single-event scenario is formally supported by an independent ancestral character reconstruction of host state, which localised a single internal *animal* $\rightarrow$ *human* transition on the branch leading to the (L6, L9, L10) crown and recovered the three re-humanised crowns as derived from a common already-human ancestor, with no internal transition inferred within (L6, L9, L10) and the four other reverse transitions confined to tip-terminal Reservoir-clade spillovers (section 3.8). A fifth corroborating line, the convergent struc-

tural losses at the *pks2*–*pks3* RDpan locus within L6.1.2 arising three times independently without coalescing into a new phylogenetic clade (section 3.9), is consistent with the same passive regime. The branch decomposition of section 3.3 further refines the geometry of this signature: only 23 of the 49 SPDIs are strictly fixed at the (L6, L9, L10) crown, and a substantial additional set of 71 strict synapomorphies separates L10 from the (L6, L9) pair. The L6L9:L6L10:L9L10 ratio of 71:1:0 favours an internal topology (L10, (L6, L9)) in which L10 diverges first, and reduces the genuine ESX-1 perturbation at the stem to a single missense in *eccE1* (with the other ESX changes either confined to the post-L10 (L6, L9) branch or appearing as L10-only apomorphies inflated by the $n = 2$ sample; fig. S4). Far from weakening the opportunistic interpretation, this stratification reinforces it: the genuine re-humanisation event mobilises a small set of fixations (three high-impact pseudogenisations and approximately twenty further substitutions) on which subsequent lineage-specific drift then accumulates. Zero-shot variant-effect scoring of the missense set with the ESM-1v protein language model [Meier et al., 2021] reinforces a predominantly conservative drift regime. Across all 27 missense substitutions – including the six PE/PPE and lipoprotein loci that the ESM Atlas does not index, recovered here from the language-model likelihoods directly – the masked-marginal log-likelihood ratio had a median of -2.8 (mean -3.6 , range -13.0 to $+2.6$); 24 of the 27 fell in the tolerated range ($\text{LLR} > -7.5$), four of them scored as favoured over the wild-type residue. Critically, the single genuine ESX-1 stem missense, *eccE1* p.Gly346Ala, scored $\text{LLR} = +0.18$, i.e. effectively neutral, and the energy-metabolism, dormancy and DNA-repair candidates *cydB* p.Met98Ile, *phoH2* p.Ala393Val, *uvrA* p.Ala938Ser and *obg* p.Ser149Pro all fell between -2.1 and -4.4 , none reaching the strongly-disfavoured regime. Only three substitutions returned large negative scores – *Rv1230c* p.Trp208Arg (-13.0), *ftsH* p.Val344Ala (-12.0) and *nudC* p.Ala268Pro (-8.1) – isolating them as the sole candidates for a locally disruptive effect. Two of these are radical substitutions (*Rv1230c* Trp→Arg and *nudC* introducing a proline) for which a strongly negative score is expected irrespective of any *in vivo* consequence, whereas *ftsH* Val→Ala is physico-chemically conservative and its very negative score most plausibly reflects the model over-penalising a highly conserved position rather than a genuine functional change; phenotyping would be required to resolve these cases. The aggregate picture is therefore one of substitutions that are individually tolerated by the language model, consistent with conservative drift and at odds with a coordinated cryptic loss-of-function or adaptive retooling of the missense set. We interpret this convergent set of null results as an *opportunistic re-colonisation* of *H. sapiens* by a lineage that could neither erase its animal-adaptation history nor re-engineer it adaptively, but was nonetheless able to exploit the human niche through a limited set of neutral fixations and ongoing genomic plasticity. Locus-specific effects at individual genes – whether adaptive, or disruptive as flagged for *ftsH*, *Rv1230c* and *nudC* – remain possible: aggregate neutrality at the scale of the 49-SPDI signature does not preclude them, but they would require gene-level branch-site tests or direct phenotyping to be detected. The tip-dated inference of section 3.7 further locates this re-colonisation in time: under the reference Menardo 2019 clock rate, the divergence of the re-humanised branch from its Wildlife clade sister (~ 1501 BCE) and the crown TMRCA of (L6, L9, L10) at ~ 992 BCE are bracketed by a long latent interval. This conclusion – single event, pre-crown stem, finite latent gap – is robust to an order-of-magnitude variation in the clock prior (section 3.7), although the absolute calendar window of the re-humanisation slides from the early Bronze Age to Late Antiquity depending on the rate chosen, and the latent-phase argument itself is attenuated at the fast-rate extreme where stem and

crown posterior intervals begin to overlap. The opportunistic re-humanisation is therefore a one-off prehistoric event, not a recurrent phenomenon continuing to produce new human-adapted lineages from the Wildlife clade pool.

4.2 Irreversibility of animal-adaptation pseudogenes

The conservation of all 48 high-impact synapomorphies across 1 556 L6, 39 L9 and 3 L10 genomes is consistent with the irreversibility of frameshift and stop-gained events: restoration through exact back-mutation is statistically implausible at the population level, and compensatory paralogues are not available for most of the affected genes. The animal-adaptation pseudogenes must therefore be either functionally redundant in the human niche, or actively compensated at the regulatory level – a hypothesis amenable to direct experimental test via transcriptomics on L6 clinical isolates. This irreversibility does not in itself constrain the re-humanisation: loss of *pks15*, *mmpL12* and the PPE5–8 cluster does not prevent colonisation of *H. sapiens*, as demonstrated by the 1 556 L6 strains carrying all three disruptions while causing endemic human tuberculosis in The Gambia [de Jong et al., 2010, Gehre et al., 2013]. The third additive high-impact pseudogenisation, an early frameshift in *Rv0909*, is biologically informative in this respect: *Rv0909* encodes the antitoxin of the Rv0909–Rv0910 type II toxin–antitoxin module, an unusual TA pair that is unique to mycobacteria, that does not belong to any of the canonical TA families (VapBC, RelBE, MazEF, HigBA, ParDE, HicAB, MenAT) and whose chromosomal position is conserved across the genus including the heavily eroded *M. leprae* [Ramage et al., 2009]. The atypical character of this module is reflected in its domain architecture, established by hmmscan against Pfam-A under gathering-threshold cutoffs [Mistry et al., 2021, Eddy, 2011]: *Rv0909* is an MT0933-like antitoxin (Pfam PF14013) carrying no recognisable DNA-binding domain, and an exploratory ESM Atlas feature profile of the 59-residue protein is dominated by polar-face amphipathic and ALPS-like lipid-packing-sensor signals rather than by the soluble DNA-binding or protein–protein-interaction signatures of canonical TA antitoxins – an ESM-derived suggestion we treat as a hypothesis. Extending the domain analysis to the two contiguous loci of the same chromosomal segment resolves the module into a small membrane-stress cluster whose biochemistry is supported by validated Pfam assignments: the cognate toxin *Rv0910* (144 aa) carries a single helix-grip SRPBCC-type lipid-binding domain (Pfam PF10604, *Polyketide_cyc2*), whose ESM features (helix-grip lipid-binding pocket, amphipathic terminal lid, glycine-rich portal loops) are more consistent with a lipid-sensor or lipase chemistry than with the ribonuclease or peptidase chemistry of canonical type II TA toxins, while its immediate chromosomal neighbour *Rv0911* (257 aa) carries tandem glyoxalase / vicinal-oxygen-chelate (VOC) domains (Pfam PF00903, two copies), consistent with the ESM cupin/VOC β -barrel and 2-His / acidic facial-triad metal-binding features that place it in the VOC metalloenzyme family detoxifying reactive carbonyls and oxidised lipids. The contiguous *Rv0909*–*Rv0910*–*Rv0911* triad therefore reads as a chromosomal mini-cluster coordinating membrane-stress sensing (the amphipathic antitoxin), release of a lipid-targeted toxin and a downstream metabolic detoxification step, a configuration distinct from the canonical translation- or replication-targeting TA systems. Against this background, the early-codon frameshift of *Rv0909* in L6, L9 and L10 is not an isolated loss of a generic antitoxin but the breakdown of one specific node of a co-clustered membrane-stress response whose downstream detoxification arm (*Rv0911*) remains intact in the signature, which reinforces the interpretation of a tolerated regional inactivation

rather than a destabilising loss. The cognate toxin Rv0910 is intact in our 49-SPDI signature, so the early loss of the antitoxin would normally release a functional toxin and be lethal under classical TA addiction. The viability of L6, L9 and L10 therefore implies either a regional functional degeneracy of this otherwise universal mycobacterial module in the modern MTBC background, or a release of its addiction circuit in the human niche where the dormancy/persistence and stringent-response programmes that TA modules typically modulate are no longer required to be as tightly regulated as in the more stressful animal niche of the Specialist-clade ancestor. Either interpretation reinforces the passive, additive regime: the re-humanised lineages have not adaptively re-engineered this regulatory module, they have simply allowed it to break.

4.3 Epidemiological precarity of L6, L9 and L10

Despite their strictly human host range, L6, L9 and L10 display persistent geographic confinement: L6 to West Africa [Afum et al., 2025, Silva et al., 2022], L9 to the Horn of Africa [Coscolla et al., 2021], L10 to a handful of isolates from the Democratic Republic of the Congo. None has achieved the global spread of L2 or L4. This qualitative observation must however be weighed against the quantitative THD analysis of section 3.10, in which we find no evidence of reduced phylogenetic clustering in the re-humanised lineages relative to the basal human ecotype L5 (L6 vs L5 non-significant at all three bandwidths; the test has limited power given the small L5 sub-sample, $n = 5$, so we cannot reject equivalence rather than positively assert it), and in which the re-humanised lineages are significantly *more* clustered than the paraphyletic Reservoir clade ($p_{\text{Bonf.}} = 7.0 \times 10^{-4}$ for the pooled comparison). We therefore describe L6, L9 and L10 as *three contextually anthroponotic branches, precarious in census count but not in phylogenetic clustering*: once they infect a human, they transmit with a tip-density comparable to L5, but the absolute number of such human hosts they have reached – 1 556 L6 genomes, 39 L9, 3 L10 against hundreds of thousands of L2 or L4 isolates and hundreds of L5 – remains minute. We propose that this ecological restriction is the epidemiological counterpart of their additive re-humanisation: the 49-SPDI compensatory signature suffices for person-to-person transmission under specific conditions (genetic background of West African / Horn-of-Africa populations, local social structure, immune context) but does not generalise, presumably because the retained animal-clade pseudogene load leaves them less competitive than the modern human lineages in other settings. The re-humanisation is therefore better framed as *locally successful opportunism*: when the three lineages reach a human host, they transmit comparably to established human ecotypes, but their global footprint remains constrained by a narrow geographical and social niche that they have never been able to escape. A persistent mislabelling in the literature attributes the 2013 Taï chimpanzee isolate [Coscolla et al., 2013] to L6, whereas the original authors explicitly proposed the informal name *Chimpanzee Bacillus* on the basis of 881 exclusive SNPs and a novel spoligotype; under current nomenclature it falls in the animal-adapted Reservoir clade of our framework, not in L6, and should therefore not be invoked as evidence of an animal reservoir for the re-humanised lineages. We also note that the characterisation of *espK* (Rv3879c) as a pan-L6 truncation marker, an extension to the lineage scale of the original observations made on the L6 reference strain GM041182 by Bentley et al. [2012] and Bold et al. [2012], is nuanced by the internal L6.1 / L6.2 topology documented here, where only the L6.1 sub-clade carries the truncation while L6.2 retains an intact copy.

4.4 Comparison with *Orygis* and *Caprae*: the Reservoir clade reservoir pivot

It is instructive to contrast the re-humanised lineages with *M. caprae* and *M. orygis*, which show the clearest incipient human-to-human transmission in Reservoir clade. Both remain reservoir-dependent: *M. caprae* retains goats and cattle as primary hosts, and the two multi-human Spanish / German sub-clusters detected here most likely reflect imported cases followed by short-range human chains rather than independent human reservoir emergence. *M. orygis*, despite a WGS-biased human frequency of 78.9%, is still primarily an ungulate pathogen, with human cases documented among South-Asian diaspora and zoo workers [Zwyer et al., 2021]. Neither has produced the kind of fully sustained, reservoir-independent human endemicity that L6 displays in The Gambia. The re-humanisation of L6, L9 and L10 is therefore a distinct evolutionary outcome from the Reservoir clade pivot: the former is a complete host switch to *H. sapiens*, confined geographically; the latter is an ongoing partial expansion from an animal reservoir, confined phylogenetically.

4.5 Implications for MTBC phylogenomics and tuberculosis surveillance

Two implications follow. First, the MTBC phylogeny is richer than a simple human-versus-animal dichotomy: the grand clade contains three distinct host-range outcomes (pure animal, reservoir with spillover, re-humanised), all emerging from the same ancestral pseudogenisation event. Host adaptation in the MTBC is thus neither strictly unidirectional (animal $\rightarrow$ human) nor strictly reversible, but context-dependent and ecologically modulated. Second, surveillance and diagnostic pipelines that rely on RD-barcoding to separate human-adapted from animal-adapted lineages [Coll et al., 2014, Napier et al., 2020] must be aware that L6, L9 and L10 will match several animal-clade RD signatures while being strictly human pathogens. Misclassification risks are real, as the four *M. suricattae* catalogue records flagged in section 3.1 illustrate.

5 Conclusion

The re-humanisation of L6, L9 and L10 is an opportunistic re-colonisation of *Homo sapiens* by a lineage that retained the irreversible pseudogenisation load of its animal-adapted ancestor. The genomic signature of this re-colonisation consists of 49 variants exclusive to the three lineages, combining three new high-impact loss-of-function events with a set of missense modifications enriched in ESX and PE/PPE families. None of the 48 animal-clade synapomorphies is reverted, and the aggregate signature behaves as a neutral fixation by drift: $d_N/d_S = 0.89$, McDonald–Kreitman $p = 0.42$, and no ascertainment bias is required to explain the observed $N:S$ ratio. The L6.1.2 RDpan locus further illustrates this passive regime by losing the same genomic region convergently on three independent sub-clades without generating a new host-range outcome. Beyond the grand-clade dataset, we identify four *M. suricattae* catalogue isolates whose RD profile is indistinguishable from authentic meerkat references while carrying a `host=Homo sapiens` metadata: a documented curiosity rather than a first zoonosis claim, to be confirmed from primary clinical records. Taken together, L6, L9 and L10 are not a re-emergence of the

L5-like ancestral human ecotype; they are three precarious, geographically confined anthroponoses carrying a paradoxical mixture of animal-adaptation and human-colonisation features, sustained by drift rather than adaptive sweeps. The Time-scaled Haplotypic Density analysis further refines this picture: the three lineages are *precarious in census count but not in phylogenetic clustering*, transmitting with a tip-density comparable to L5 once they reach a human host, while remaining globally minute in absolute numbers. The tip-dated phylogeny resolves the outstanding temporal question in favour of a single ancestral event: the re-humanised branch diverged from the Wildlife clade around 1501 BCE, coalesced into the monophyletic (L6, L9, L10) crown around 992 BCE after a ~500-year latent phase (point-estimate gap, with overlapping 90% CIs at the stem and crown nodes), with the internal diversification of the three sampled crown lineages then unfolding during the first millennium CE. An independent ancestral character reconstruction of host state, performed with PastML on the same tip-dated tree, formally confirms this scenario by localising a single internal *animal* → *human* transition on the branch leading to the (L6, L9, L10) crown and inferring no internal transition within the three lineages. The 49-SPDI signature was thus acquired on a single pre-crown stem and inherited vertically by all three lineages, rather than accumulated convergently through three independent re-humanisations. The re-humanisation of the MTBC into *H. sapiens* is in this sense a one-off prehistoric contingency, not an on-going recurrent process.

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Supplementary Materials

Sensitivity of the L6 n L9 n L10 signature to the core frequency threshold

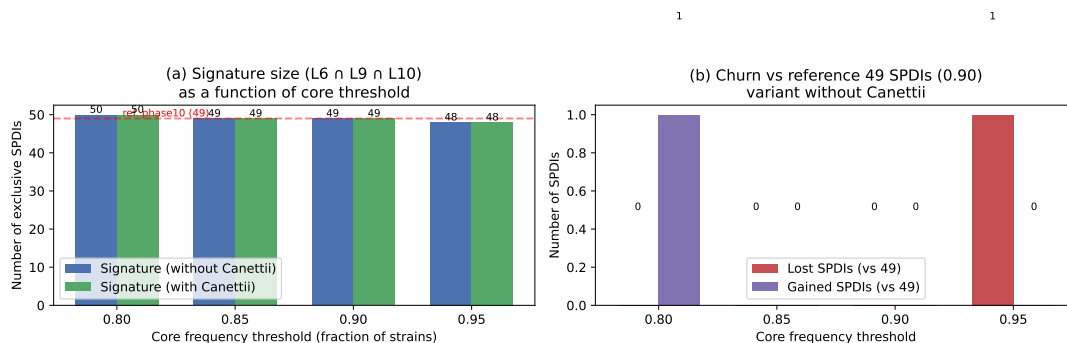


Figure S1: Sensitivity of the 49-SPDI re-humanisation signature to the core-frequency threshold. Signature sizes and overlap with the reference signature at 0.90 are plotted for thresholds 0.80, 0.85, 0.90 and 0.95, with and without additional outgroup polarisation against *M. canettii*. Forty-eight of the 49 reference SPDI variants form an invariant core across all four thresholds; a single in-frame insertion in *Rv0230c* (*php*) is lost at the most stringent 0.95 cut-off, and a single missense variant in *Rv1266c* (*pknH*) is gained at the most permissive 0.80 cut-off.

Table S1: Sensitivity of the re-humanisation signature to the core-frequency threshold. Columns report the size of the core of each re-humanised lineage, the size of their intersection, the size of the exclusive signature with and without Canettii polarisation, and the number of reference-49 SPDI variants retained. Full table provided as Supplementary Table S1.

Thr.	core L6	core L9	core L10	Inter.	Sig. A	Sig. B	Ref. A	Ref. B
0.80	1 813	1 852	2 213	1 393	50	50	49/49	49/49
0.85	1 786	1 839	2 213	1 376	49	49	49/49	49/49
0.90	1 759	1 812	2 213	1 357	49	49	49/49	49/49
0.95	1 710	1 770	2 213	1 310	48	48	48/49	48/49

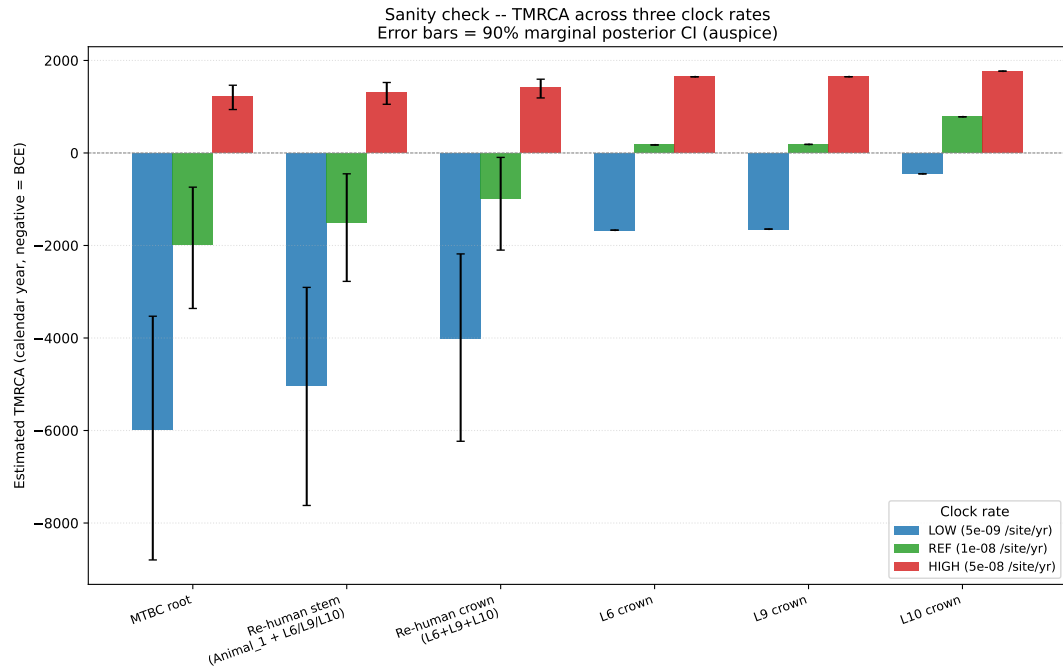


Figure S2: Sensitivity of the tip-dated TreeTime analysis to the clock rate. TMRCA point estimates for the MTBC root, the Wildlife clade + (L6, L9, L10) stem, the (L6, L9, L10) crown and the three lineage crowns are shown for three strict-clock rates spanning an order of magnitude: low (5×10^{-9} sub/site/yr, aDNA-calibrated), reference (1×10^{-8}) and high (5×10^{-8} , short-term). Absolute dates scale inversely with the rate; the monophyly of (L6, L9, L10) and the ordering stem < crown < within-lineage crowns are preserved at all rates.

Table S2: Sensitivity of the tip-dated TMRCA estimates to the clock rate prior. Dates are in BCE/CE; the stem-to-crown gap is in years. H1 refers to the single-ancestral-re-humanisation hypothesis, which requires monophyly of (L6, L9, L10) together with a positive stem-to-crown gap. Full table provided as Supplementary Table S2.

Rate	MTBC root	Stem	Crown	L6	L9	L10	Gap (yr)	Monophyl.	H1
low	5977 BCE	5018 BCE	4002 BCE	1669 BCE	1646 BCE	451 BCE	1016	yes	yes
ref.	1980 BCE	1501 BCE	992 BCE	174 CE	185 CE	782 CE	510	yes	yes
high	1217 CE	1313 CE	1414 CE	1648 CE	1649 CE	1769 CE	102	yes	yes

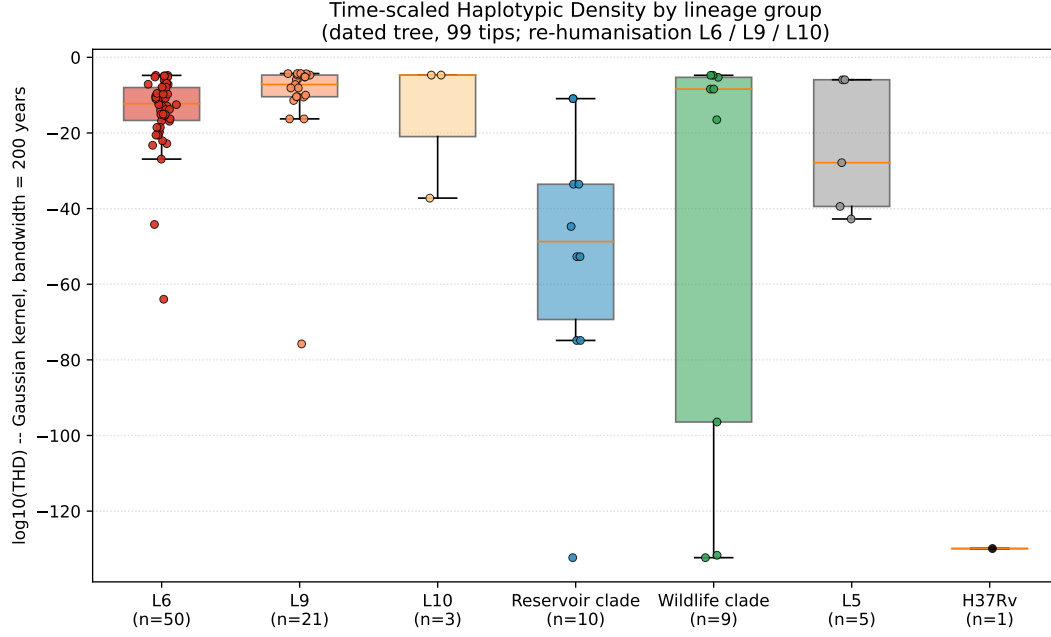


Figure S3: Distribution of Time-scaled Haplotypic Density (THD) values per MTBC group on the 99-taxon tip-dated tree, at the central bandwidth of 200 years. Each violin shows the per-tip THD within a group (re-humanised lineages L6, L9, L10; pooled re-humanised ReHuman; Reservoir clade; Wildlife clade; L5). A companion Supplementary Figure projects the same THD values onto the tip-dated tree as a colour gradient. Bandwidth sensitivity (50 y, 200 y, 500 y) preserves the qualitative ranking.

Table S3: Per-group THD summary and pre-declared Mann–Whitney contrasts (Bonferroni-corrected, bandwidth = 200 years). Full outputs for the three bandwidths are provided as Supplementary Table S3.

Contrast	Median A	Median B	$p_{\text{Bonf.}}$
L6 vs L5	6.63×10^{-13}	1.39×10^{-28}	1.00 (ns)
L6 vs L9	6.63×10^{-13}	6.45×10^{-8}	2.4×10^{-2}
L6 vs L10	6.63×10^{-13}	2.03×10^{-5}	1.00 (ns)
L6 vs Reservoir clade	6.63×10^{-13}	9.15×10^{-46}	2.0×10^{-3}
L6 vs Wildlife clade	6.63×10^{-13}	4.00×10^{-9}	1.00 (ns)
L9 vs L5	6.45×10^{-8}	1.39×10^{-28}	0.87 (ns)
L9 vs Reservoir clade	6.45×10^{-8}	9.15×10^{-46}	1.7×10^{-3}
L9 vs Wildlife clade	6.45×10^{-8}	4.00×10^{-9}	1.00 (ns)
ReHuman vs Reservoir clade	3.90×10^{-11}	9.15×10^{-46}	7.0×10^{-4}
ReHuman vs Wildlife clade	3.90×10^{-11}	4.00×10^{-9}	1.00 (ns)
ReHuman vs L5	3.90×10^{-11}	1.39×10^{-28}	1.00 (ns)

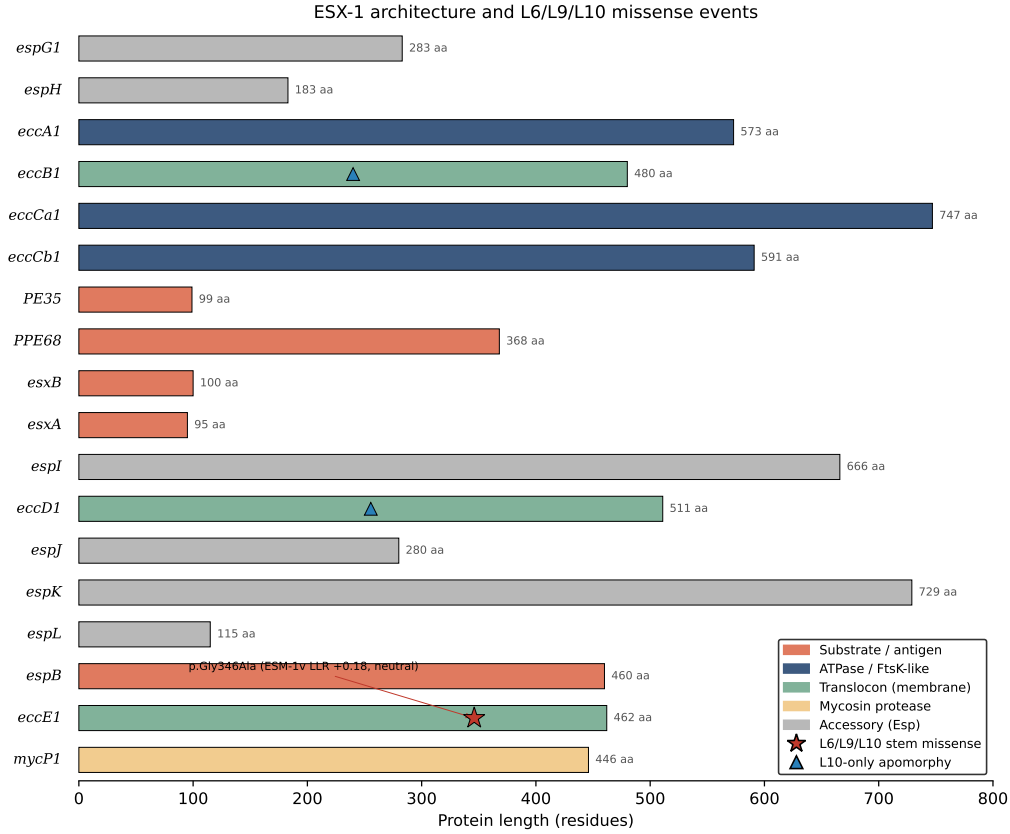


Figure S4: Architecture of the ESX-1 type VII secretion system and position of the L6/L9/L10 missense events recovered by the 49-SPDI signature and its branch decomposition (section 3.2, section 3.3). Each bar represents one H37Rv ESX-1 gene, scaled to its protein length and coloured by functional category (substrate / antigen, ATPase / FtsK-like, translocon / membrane, mycosin protease, accessory Esp). The red star marks the only stem missense, *eccE1* p.Gly346Ala, which the ESM-1v language model scores as effectively neutral (LLR = +0.18; section 4.1). The blue triangles flag the two L10-only ESX-1 apomorphies (*eccB1*, *eccD1*) inflated by the $n = 2$ L10 sample. The three (L6, L9) post-L10 ESX missense changes (*esxH*, *eccD5*, *eccA2*) discussed in section 4.1 fall in ESX-3, ESX-5 and ESX-2 paralogues respectively and are therefore not on this ESX-1 map.

Table S4: Host-state transitions inferred by Fitch parsimony from the maximum-likelihood ancestral states reconstructed by PastML on the 99-taxon tip-dated tree (section 2.12, section 3.8). Branch type indicates whether the transition is internal (subtending a clade of ≥ 2 tips) or tip-terminal (single-tip spillover). Full per-node posterior probabilities are provided as Supplementary Table S4 and the interactive PastML tree as a Supplementary HTML file.

From	To	Count	Branch type / location
human	animal	1	internal: MTBC root $\rightarrow$ grand-clade ancestor (Reservoir clade + L5 + L6/9/10)
animal	human	1	internal: stem (Wildlife clade, (L6+L9+L10)) $\rightarrow$ crown (L6+L9+L10)
animal	human	4	tip-terminal: Pinnipedii, Microti, Orygis, Caprae spillovers in the Reservoir clade
animal	human	0	internal within (L6, L9, L10)

Table S5: Strict synapomorphies of the L6.1 intermediate node: 68 variants present at $\geq 90\%$ in both L6.1.1 and L6.1.2 and absent at $< 10\%$ from L6.2, inferred on the 1556-strain L6 cohort of this study (936 L6.1.1, 256 L6.1.2 and 364 L6.2 ; section 2.1) and absent from all outgroup clades at the same threshold. HGVS.p annotations are derived from snpEff against H37Rv (NC_000962.3).

Position	SPDI	Gene	Effect	HGVS.p
<i>HIGH-impact (6)</i>				
1265911	NC_000962.3:1265911:GA:G	Rv1138c	frameshift	p.Pro193fs
1962503	NC_000962.3:1962503:GC:G	Rv1736c	frameshift	p.Gly561fs
2448401	NC_000962.3:2448401:C:A	Rv2187	stop_gained	p.Tyr81*
3486711	NC_000962.3:3486711:C:CG	Rv3121	frameshift	p.Pro69fs
4179387	NC_000962.3:4179387:GTGGTCGGCGTCGGCGTGCCAGGCACCGACACCA:G	Rv3729	frameshift (del 34 bp)	p.Trp369fs
4359354	NC_000962.3:4359354:GT:A	Rv3879c (<i>espK</i>)	frameshift	p.Thr143fs
<i>MODERATE-impact missense (37)</i>				
8492	NC_000962.3:8492	Rv0006	missense	p.Leu398Phe
76146	NC_000962.3:76146	Rv0068	missense	p.Gln283Glu
147649	NC_000962.3:147649	Rv0120c	missense	p.Glu41Gly
208402	NC_000962.3:208402	Rv0176	missense	p.Pro318Ser
225415	NC_000962.3:225415	Rv0193c	missense	p.Trp386Arg
360584	NC_000962.3:360584	Rv0296c	missense	p.Glu191Lys
405273	NC_000962.3:405273	Rv0338c	missense	p.Leu190Ile
425213	NC_000962.3:425213	Rv0355c	missense	p.Gly3156Ser
476581	NC_000962.3:476581	Rv0397A	missense	p.Trp63Cys
598243	NC_000962.3:598243	Rv0507	missense	p.Thr349Ile
654602	NC_000962.3:654602	Rv0563	missense	p.Glu242Gly
705987	NC_000962.3:705987	Rv0611c	missense	p.Glu65Asp
816731	NC_000962.3:816731	Rv0724	missense	p.Ser357Asn
867744	NC_000962.3:867744	Rv0774c	missense	p.Thr203Ala
919381	NC_000962.3:919381	Rv0825c	missense	p.TyrPhe57CysLeu
1347263	NC_000962.3:1347263	Rv1204c	missense	p.Thr454Met
1406684	NC_000962.3:1406684	Rv1258c	missense	p.Val219Ala
1416632	NC_000962.3:1416632	Rv1267c	missense	p.Leu239Val
1593651	NC_000962.3:1593651	Rv1419	missense	p.Ala50Thr
1620177	NC_000962.3:1620177	Rv1442	missense	p.Met130Pro
1774115	NC_000962.3:1774115	Rv1566c	missense	p.Thr169Pro
1837167	NC_000962.3:1837167	Rv1633	missense	p.Ala32Ser
2122379	NC_000962.3:2122379	Rv1872c	missense	p.Leu258Val
2372950	NC_000962.3:2372950	Rv2113	missense	p.Gly108Arg
2428952	NC_000962.3:2428952	Rv2165c	missense	p.Gly158Asp
2593620	NC_000962.3:2593620	Rv2320c	missense	p.Ala178Glu
2618061	NC_000962.3:2618061	Rv2340c	missense	p.Gly283Arg
2848799	NC_000962.3:2848799	Rv2524c	missense	p.Ala178Val
2900966	NC_000962.3:2900966	Rv2577	missense	p.Gly17Glu
2954317	NC_000962.3:2954317	Rv2627c	missense	p.Thr144Met
3054080	NC_000962.3:3054080	Rv2741	missense	p.His58Tyr
3149677	NC_000962.3:3149677	Rv2842c	missense	p.Arg100His
3400475	NC_000962.3:3400475	Rv3040c	missense	p.Ala195Ser
3603177	NC_000962.3:3603177	Rv3226c	missense	p.His49Tyr
3894776	NC_000962.3:3894776	Rv3478	missense	p.Leu118Val
4128151	NC_000962.3:4128151	Rv3685c	missense	p.Val192Ile
4141284	NC_000962.3:4141284	Rv3698	missense	p.Arg265Cys
4375096	NC_000962.3:4375096	Rv3892c	missense	p.Ala196Val
4375970	NC_000962.3:4375970	Rv3893c	missense	p.Ala9Thr
<i>LOW-impact synonymous (18)</i>				
144344	NC_000962.3:144344	Rv0119	synonymous	p.Arg99Arg
162621	NC_000962.3:162621	Rv0134	synonymous	p.Val284Val
168528	NC_000962.3:168528	Rv0142	synonymous	p.Pro273Pro
538761	NC_000962.3:538761	Rv0450c	synonymous	p.Leu910Leu
698967	NC_000962.3:698967	Rv0601c	synonymous	p.Gly9Gly
772595	NC_000962.3:772595	Rv0672	synonymous	p.Arg371Arg
866447	NC_000962.3:866447	Rv0773c	synonymous	p.Thr314Thr
2501400	NC_000962.3:2501400	Rv2227	synonymous	p.Pro157Pro
2770010	NC_000962.3:2770010	Rv2467	synonymous	p.Ala342Ala
2912814	NC_000962.3:2912814	Rv2586c	synonymous	p.Gly399Gly
3255168	NC_000962.3:3255168	Rv2932	synonymous	p.Ala1366Ala
3463723	NC_000962.3:3463723	Rv3094c	synonymous	p.Ala56Ala
4195896	NC_000962.3:4195896	Rv3745c	synonymous	p.Leu68Leu
4228996	NC_000962.3:4228996	Rv3782	synonymous	p.Cys217Cys
4303553	NC_000962.3:4303553	Rv3829c	synonymous	p.Gly485Gly
4339879	NC_000962.3:4339879	Rv3863	synonymous	p.Ala344Ala
4364322	NC_000962.3:4364322	Rv3883c	synonymous	p.Asp145Asp
4372274	NC_000962.3:4372274	Rv3888c	synonymous	p.Gly144Gly
<i>Intergenic / non-coding (7)</i>				
1384254	NC_000962.3:1384254	—	intergenic	—
2053438	NC_000962.3:2053438	—	intergenic	—
3743548	NC_000962.3:3743548	—	intergenic	—
4127008	NC_000962.3:4127008	—	intergenic	—

3 additional positions with dual snpEff annotations (intergenic / intragenic_variant) included in the count.